

BLUESTOCK MUTUAL FUND ANALYTICS PLATFORM

An End-to-End Data Analytics Capstone Project

Final Project Report

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Technologies Used

- Python
- Pandas
- NumPy
- SQLite
- SQL
- Plotly
- Matplotlib
- Seaborn
- Power BI

1. Introduction

1.1 Background

The Indian mutual fund industry has experienced remarkable growth over the past decade, driven by increasing financial awareness, digital investment platforms, and the rising popularity of Systematic Investment Plans (SIPs). With thousands of mutual fund schemes offered by multiple Asset Management Companies (AMCs), investors face challenges in comparing fund performance, understanding risk, and making informed investment decisions.

Data analytics provides an effective approach to transform raw financial data into actionable insights. By integrating historical NAV data, Assets Under Management (AUM), investor transactions, benchmark indices, and portfolio holdings, meaningful analyses can be performed to evaluate fund performance, identify market trends, measure investment risk, and understand investor behaviour.

This project demonstrates a complete data analytics workflow by developing an end-to-end mutual fund analytics platform. The solution incorporates data cleaning, database design, exploratory data analysis, quantitative performance metrics, advanced risk analytics, and interactive business intelligence dashboards to support data-driven investment decisions.

1.2 Project Overview

The Bluestock Mutual Fund Analytics Platform was developed to demonstrate the complete lifecycle of a modern data analytics project. Beginning with raw datasets, the project applies ETL (Extract, Transform, Load) processes to clean and standardize the data before storing it in a relational SQLite database. SQL queries are then used to retrieve and analyze the data efficiently.

Building upon the prepared dataset, extensive exploratory data analysis (EDA) was performed to uncover trends in fund performance, SIP inflows, AUM growth, investor demographics, and portfolio allocation. Advanced financial metrics, including Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), Conditional Value at Risk (CVaR), and rolling performance indicators, were implemented to evaluate mutual fund performance from a risk-adjusted perspective.

Finally, an interactive Power BI dashboard was developed to provide users with dynamic visualizations, drill-through navigation, KPI cards, and filtering capabilities, enabling intuitive exploration of mutual fund performance and industry trends.

2. Problem Statement

The rapidly expanding mutual fund industry generates large volumes of financial and investor-related data. Although this information is publicly available, it is often distributed across multiple sources and formats, making comprehensive analysis difficult for investors, analysts, and financial institutions.

Traditional methods of comparing mutual funds typically rely on a limited set of performance indicators, such as historical returns or fund ratings, without considering risk-adjusted returns, downside risk, investor behaviour, or portfolio diversification. Furthermore, the absence of integrated dashboards limits the ability to perform interactive analysis across multiple dimensions, including fund category, fund house, investment plan, geographic distribution, and benchmark performance.

The objective of this project is to address these challenges by developing an integrated analytics platform capable of cleaning, organizing, analyzing, and visualizing mutual fund data while generating meaningful business insights to support informed investment decisions.

3. Project Objectives

The primary objectives of this project are:

- Develop an automated ETL pipeline for cleaning and transforming raw mutual fund datasets.
- Design and implement a normalized SQLite database for efficient storage and querying.
- Perform comprehensive Exploratory Data Analysis (EDA) to identify industry trends and investment patterns.
- Calculate key financial performance metrics, including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Value at Risk.
- Analyze investor behaviour through demographic, geographic, cohort, and SIP continuity analyses.
- Build an interactive Power BI dashboard featuring KPIs, slicers, drill-through navigation, and benchmark comparisons.

- Generate business insights that support mutual fund evaluation and investment decision-making.

4. Scope of the Project

This project focuses on analyzing historical mutual fund data collected for selected Indian mutual fund schemes between 2022 and 2025. The analysis includes historical NAV movements, Assets Under Management (AUM), SIP inflows, investor transaction records, benchmark indices, and portfolio holdings.

The scope encompasses data engineering, statistical analysis, financial performance evaluation, and business intelligence visualization. While the project provides a comprehensive analytical framework for mutual fund analysis, it does not aim to predict future market prices or provide personalized financial advice. Instead, it serves as a decision-support system by presenting historical trends, quantitative risk measures, and interactive dashboards that assist users in evaluating mutual fund performance from multiple perspectives.

5. Dataset Description

5.1 Dataset Overview

The Bluestock Mutual Fund Analytics Platform utilizes multiple datasets representing different aspects of the Indian mutual fund ecosystem. These datasets include mutual fund master information, historical NAV prices, Assets Under Management (AUM), SIP inflows, investor transactions, portfolio holdings, and benchmark indices. Together, they provide a comprehensive foundation for financial analysis, risk assessment, investor behavior analysis, and business intelligence reporting.

Prior to analysis, all datasets underwent data cleaning and standardization to ensure consistency, remove duplicates, handle missing values, and maintain data integrity.

Table 5.1 – Dataset Summary

Dataset	Records	Description
Fund Master	40	Mutual fund scheme metadata including fund house, category, plan, and risk grade
NAV History	46,000	Daily Net Asset Value (NAV) history from 2022–2026
AUM by Fund House	90	Assets Under Management across leading AMCs
Monthly SIP Inflows	48	Monthly SIP collection statistics
Category Inflows	144	Monthly inflows across mutual fund categories
Industry Folio Count	21	Growth in investor folios over time
Scheme Performance	40	Fund performance and risk metrics
Investor Transactions	32,778	Individual investment transactions
Portfolio Holdings	322	Sector-wise portfolio allocation
Benchmark Indices	8,050	Historical benchmark index values

5.2 Data Sources

The project integrates publicly available financial datasets related to the Indian mutual fund industry. These datasets represent historical market performance, investor activity, benchmark indices, and fund characteristics, providing a comprehensive basis for analytics and visualization.

6. Technology Stack

A modern analytics stack was selected to support the complete data lifecycle, including ETL, storage, statistical analysis, visualization, and business intelligence.

Table 6.1 – Technology Stack

Technology	Purpose
Python	ETL pipeline and analytics
Pandas	Data cleaning and manipulation
NumPy	Numerical computations
SQLite	Relational database
SQL	Data querying
Matplotlib	Statistical visualizations
Seaborn	Advanced visualizations
Plotly	Interactive charts
Jupyter Notebook	Exploratory analysis
Power BI	Interactive dashboard
Git & GitHub	Version control

Technology Selection

Python was selected because of its mature ecosystem for financial analytics and data science. Pandas and NumPy enabled efficient preprocessing and numerical analysis, while SQLite provided a lightweight relational database suitable for local development. Power BI was used to build interactive dashboards with drill-through, slicers, and KPI cards, allowing business users to explore fund performance dynamically. Choosing technologies with strong community support and appropriate capabilities is a common expectation in capstone projects.

7. Data Cleaning and ETL Pipeline

7.1 ETL Overview

An automated ETL (Extract, Transform, Load) pipeline was developed to transform raw datasets into analysis-ready data. The pipeline ensures consistency, reproducibility, and data quality before loading the processed datasets into the SQLite database.

The ETL process consisted of three primary stages:

- **Extract:** Import raw CSV datasets.
- **Transform:** Perform data cleaning, type conversion, duplicate removal, missing value handling, and column standardization.
- **Load:** Export cleaned datasets and populate the SQLite database.

7.2 Data Cleaning Operations

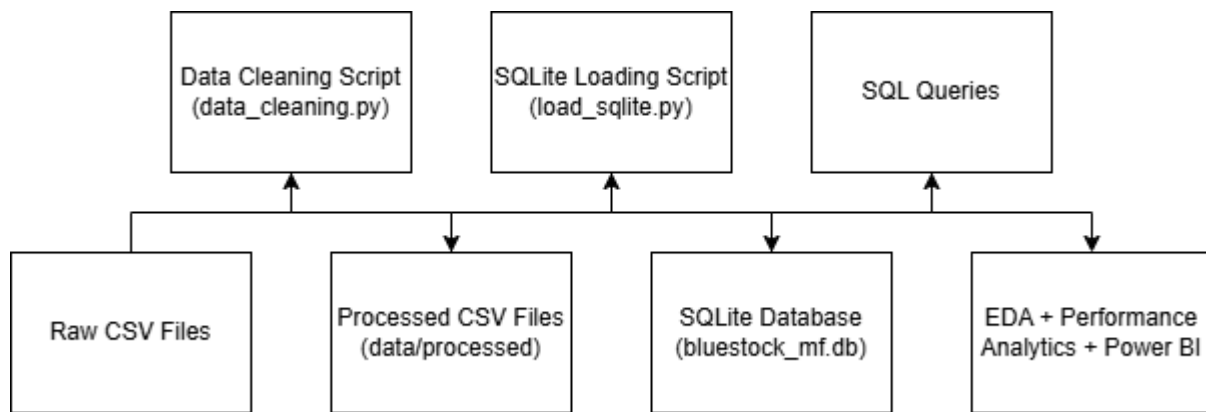
- The following preprocessing steps were performed across the datasets:
- Removed duplicate records.
- Standardized column names.
- Converted date columns to YYYY-MM-DD format.
- Converted numerical fields to appropriate numeric data types.
- Handled missing and invalid values.
- Verified primary key uniqueness.
- Exported cleaned datasets into the data/processed directory.

7.3 SQLite Loading

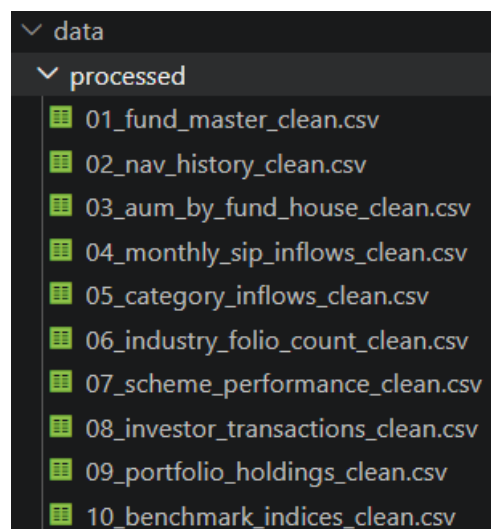
After preprocessing, the cleaned datasets were loaded into SQLite using an automated Python script. The loading process included:

- Automatic table creation.
- Bulk insertion of cleaned data.
- Row-count verification.
- Schema validation.

This ensured that all analytical notebooks and SQL queries operated on a consistent and validated database.



(ETL Pipeline for Mutual Fund Data Processing.)



(Successful loading of cleaned datasets into the SQLite database.)

8. Database Design

8.1 Database Overview

A relational SQLite database was designed to organize and manage the processed mutual fund data efficiently. The schema separates descriptive information from transactional and time-series data, supporting efficient querying, analytical computations, and dashboard reporting.

The database follows a star-like analytical structure, where the **Fund Master** table acts as the primary reference entity, linked to multiple analytical tables using the **AMFI Code** as the key identifier.

8.2 Database Tables

The database contains the following primary tables:

Table	Purpose
dim_fund	Master information for mutual fund schemes
fact_nav	Historical daily NAV values
fact_aum	Assets Under Management by fund house
monthly_sip_inflows	Monthly SIP collection
category_inflows	Category-wise net inflows
industry_folio_count	Industry folio statistics
fact_performance	Performance metrics for each scheme
fact_transactions	Investor transaction history
portfolio_holdings	Sector allocation of schemes
benchmark_indices	Historical benchmark index values

8.3 Database Relationships

The principal relationships include:

- **AMFI Code** linking fund metadata with NAV, performance, transactions, and portfolio holdings.
- **Date** linking time-series datasets such as NAV, SIP inflows, AUM, and benchmark indices.
- One-to-many relationships between the fund dimension and multiple fact tables.

These relationships support efficient analytical queries and enable consistent filtering across notebooks and Power BI dashboards.

Name	Type	Schema
Tables (12)		
benchmark_indices		CREATE TABLE benchmark_indices (date TEXT, index_name TEXT, close_value FLOAT)
category_inflows		CREATE TABLE category_inflows (month TEXT, category TEXT, net_inflow_crore FLOAT)
dim_date		CREATE TABLE dim_date (date_key INTEGER PRIMARY KEY, full_date DATE, year INTEGER, quarter INTEGER, month INTEGER)
dim_fund		CREATE TABLE dim_fund (amfi_code BIGINT, fund_house TEXT, scheme_name TEXT, category TEXT, sub_category TEXT)
fact_aum		CREATE TABLE fact_aum (date TEXT, fund_house TEXT, aum_lakh_crore FLOAT, aum_crore BIGINT, num_schemes BIGINT)
fact_nav		CREATE TABLE fact_nav (amfi_code BIGINT, date TEXT, nav FLOAT)
fact_performance		CREATE TABLE fact_performance (amfi_code BIGINT, scheme_name TEXT, fund_house TEXT, category TEXT, "plan" TEXT)
fact_transactions		CREATE TABLE fact_transactions (investor_id TEXT, transaction_date TEXT, amfi_code BIGINT, transaction_type TEXT)
industry_folio_count		CREATE TABLE industry_folio_count (month TEXT, total_folios_crore FLOAT, equity_folios_crore FLOAT, debt_folios_crore FLOAT)
monthly_sip_inflows		CREATE TABLE monthly_sip_inflows (month TEXT, sip_inflow_crore BIGINT, active_sip_accounts_crore FLOAT, new_sip_accounts_crore FLOAT)
portfolio_holdings		CREATE TABLE portfolio_holdings (amfi_code BIGINT, stock_symbol TEXT, stock_name TEXT, sector TEXT, weight_crore FLOAT)
sqlite_sequence		CREATE TABLE sqlite_sequence(name,seq)
Indices (0)		
Views (0)		
Triggers (0)		

(SQLite database schema illustrating relationships between analytical tables.)

SQL 1*					
1	SELECT				
2	scheme_name,				
3	fund_house,				
4	category,				
5	return_3yr_pct,				
6	sharpe_ratio				
7	FROM fact_performance				
8	ORDER BY sharpe_ratio DESC				
9	LIMIT 10;				
	scheme_name	fund_house	category	return_3yr_pct	sharpe_ratio
1	ICICI Pru Liquid Fund - Regular - ...	ICICI Prudential MF	Liquid	7.68	7.68
2	Kotak Liquid Fund - Regular - Growth	Kotak Mahindra MF	Liquid	6.18	6.18
3	ABSL Liquid Fund - Regular - Growth	Aditya Birla Sun Life MF	Liquid	5.14	5.14
4	HDFC Short Term Debt Fund - Regular ...	HDFC Mutual Fund	Short Duration	7.37	1.84
5	SBI Magnum Gilt Fund - Regular Plan ...	SBI Mutual Fund	Gilt	6.07	1.52
6	Nippon India Gilt Securities Fund - ...	Nippon India MF	Gilt	5.31	1.33
7	HDFC Top 100 Fund - Regular Plan - ...	HDFC Mutual Fund	Large Cap	14.84	1.06
8	Mirae Asset Large Cap Fund - Regular...	Mirae Asset MF	Large Cap	14.81	1.06
9	ICICI Pru Bluechip Fund - Direct - ...	ICICI Prudential MF	Large Cap	14.41	1.03
10	Nippon India Large Cap Fund - Regula...	Nippon India MF	Large Cap	14.0	1.0

(Sample SQL query demonstrating analytical retrieval from the SQLite database.)

8.4 Database Validation

- After loading, validation checks confirmed:
- All expected tables were created successfully.
- Data types matched the schema definition.
- Row counts matched the processed datasets.
- SQL queries returned the expected analytical results without integrity issues.

9. Exploratory Data Analysis (EDA)

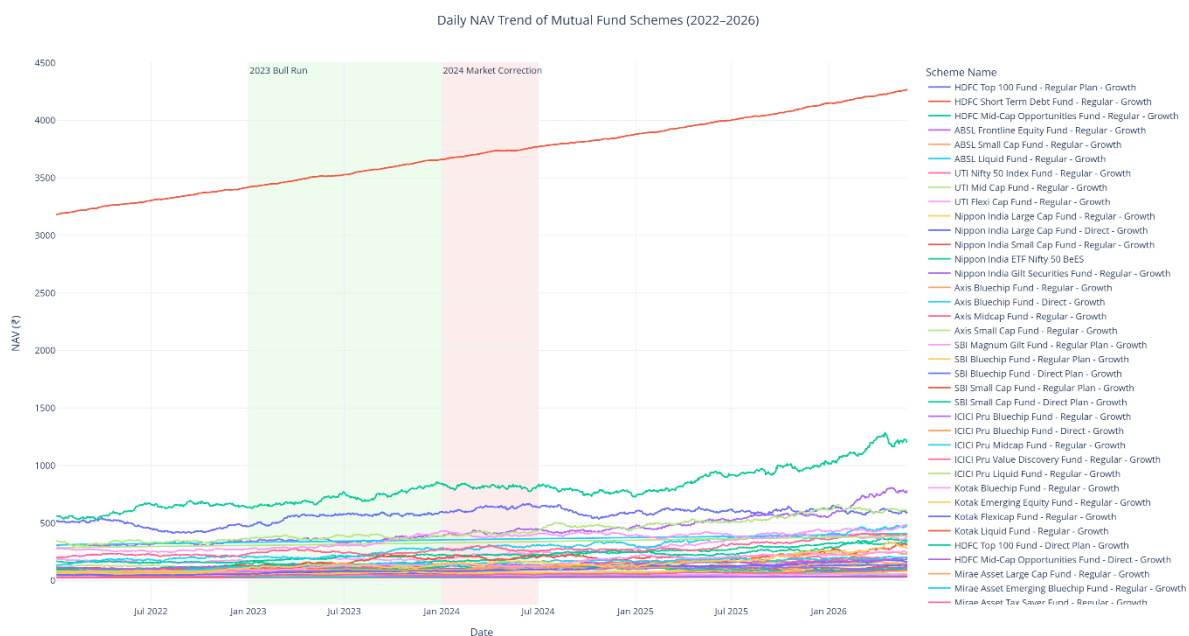
9.1 Overview

Exploratory Data Analysis (EDA) was performed to understand the structure, trends, and relationships within the mutual fund datasets before conducting advanced financial analysis. Multiple statistical and visualization techniques were used to examine historical NAV movements, Assets Under Management (AUM), SIP inflows, investor demographics, portfolio allocation, and benchmark performance.

The objective of the EDA was to identify meaningful trends, detect anomalies, understand investor behavior, and generate insights that support informed investment decisions.

9.2 Daily NAV Trend Analysis

Daily Net Asset Value (NAV) data for all selected mutual fund schemes was analyzed over the period from 2022 to 2026. The analysis highlighted long-term growth patterns, market corrections, and differences in fund performance.

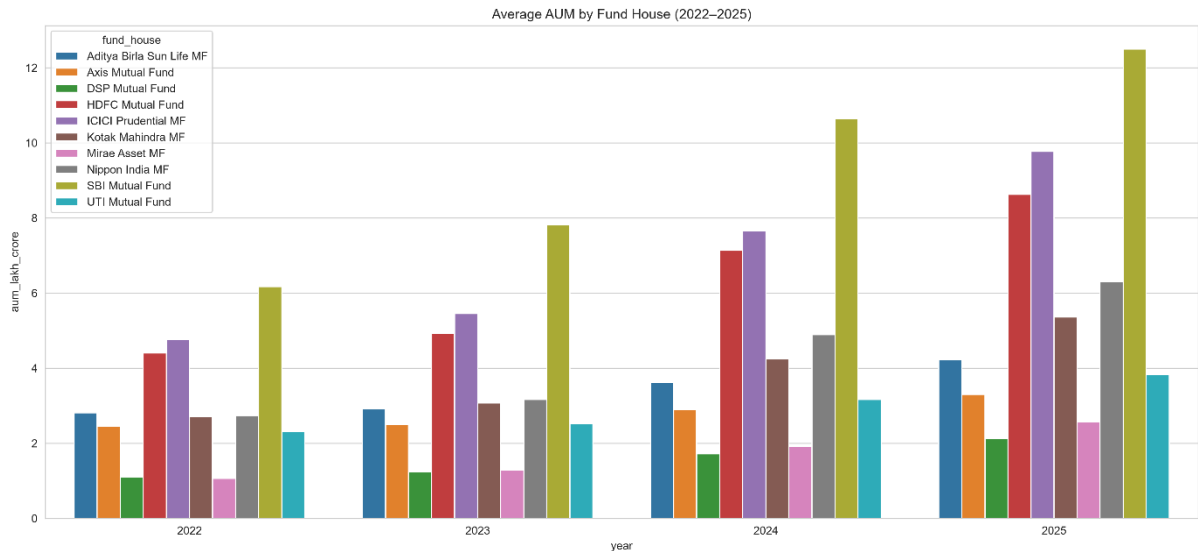


Observation

- Most equity-oriented schemes demonstrated consistent long-term appreciation.
- A noticeable upward trend was observed during the 2023 market recovery.
- Temporary declines during market corrections were reflected across most schemes before resuming upward movement.

9.3 Assets Under Management (AUM) Analysis

AUM growth was analyzed across leading Asset Management Companies to evaluate market dominance and industry expansion.

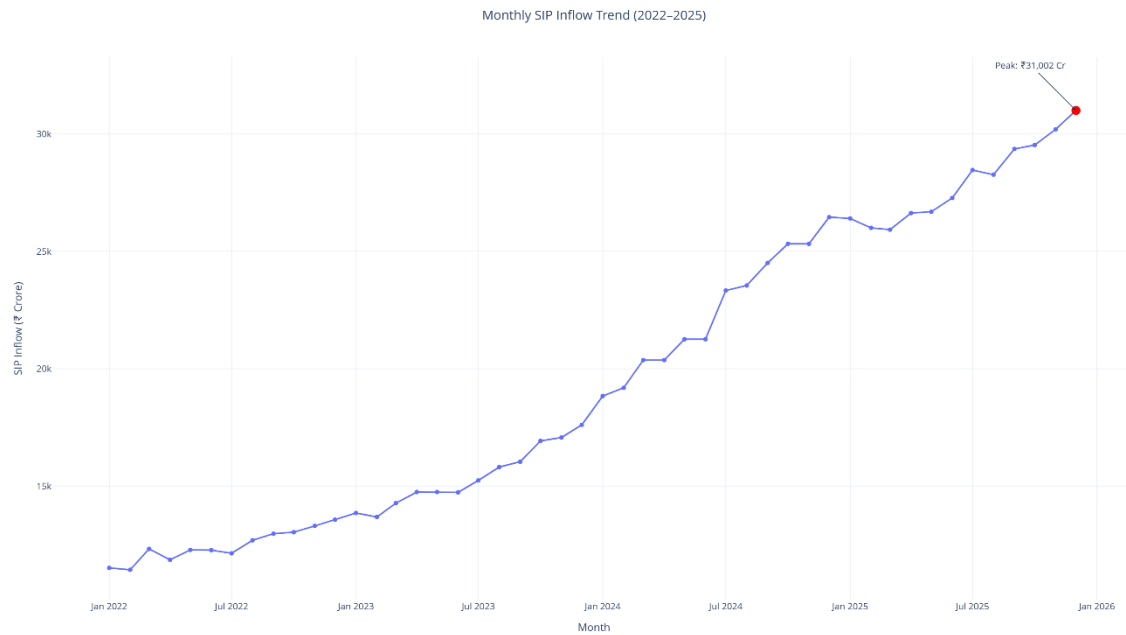


Observation

- SBI Mutual Fund maintained the highest Assets Under Management throughout the study period.
- The industry experienced consistent AUM growth, reflecting increased investor participation.
- Large AMCs continued to dominate market share.

9.4 SIP Inflow Analysis

Monthly Systematic Investment Plan (SIP) collections were examined to understand investor participation trends.

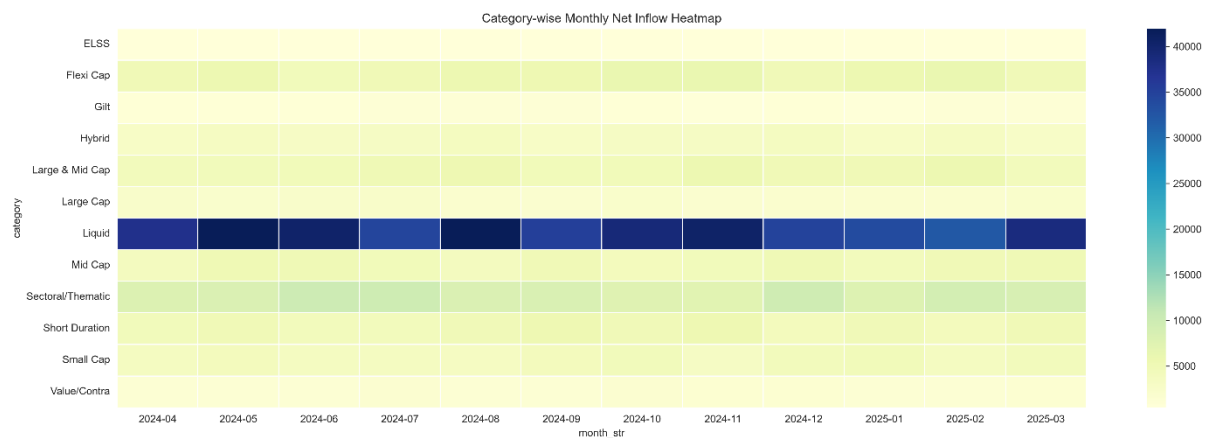


Observation

- SIP inflows increased steadily during the analysis period.
- Monthly investments reached record levels toward the end of 2025.
- The trend indicates increasing investor confidence in disciplined long-term investing.

9.5 Category-wise Net Inflows

Net inflows across mutual fund categories were analyzed using a heatmap.

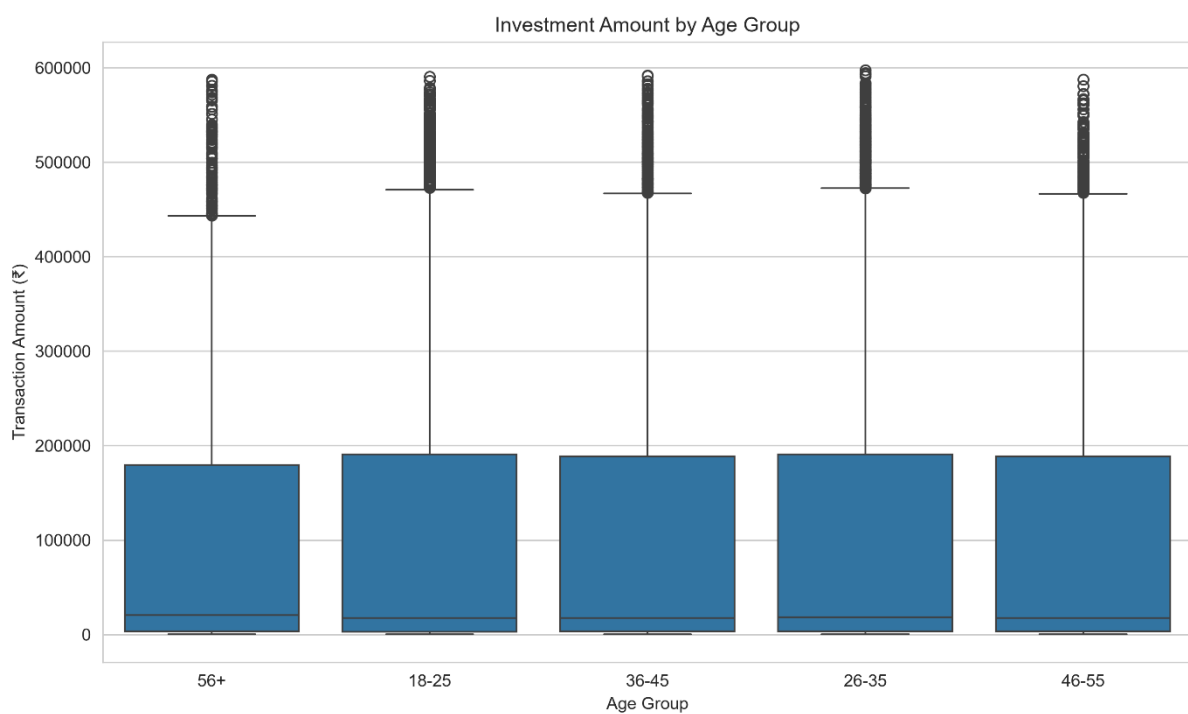


Observation

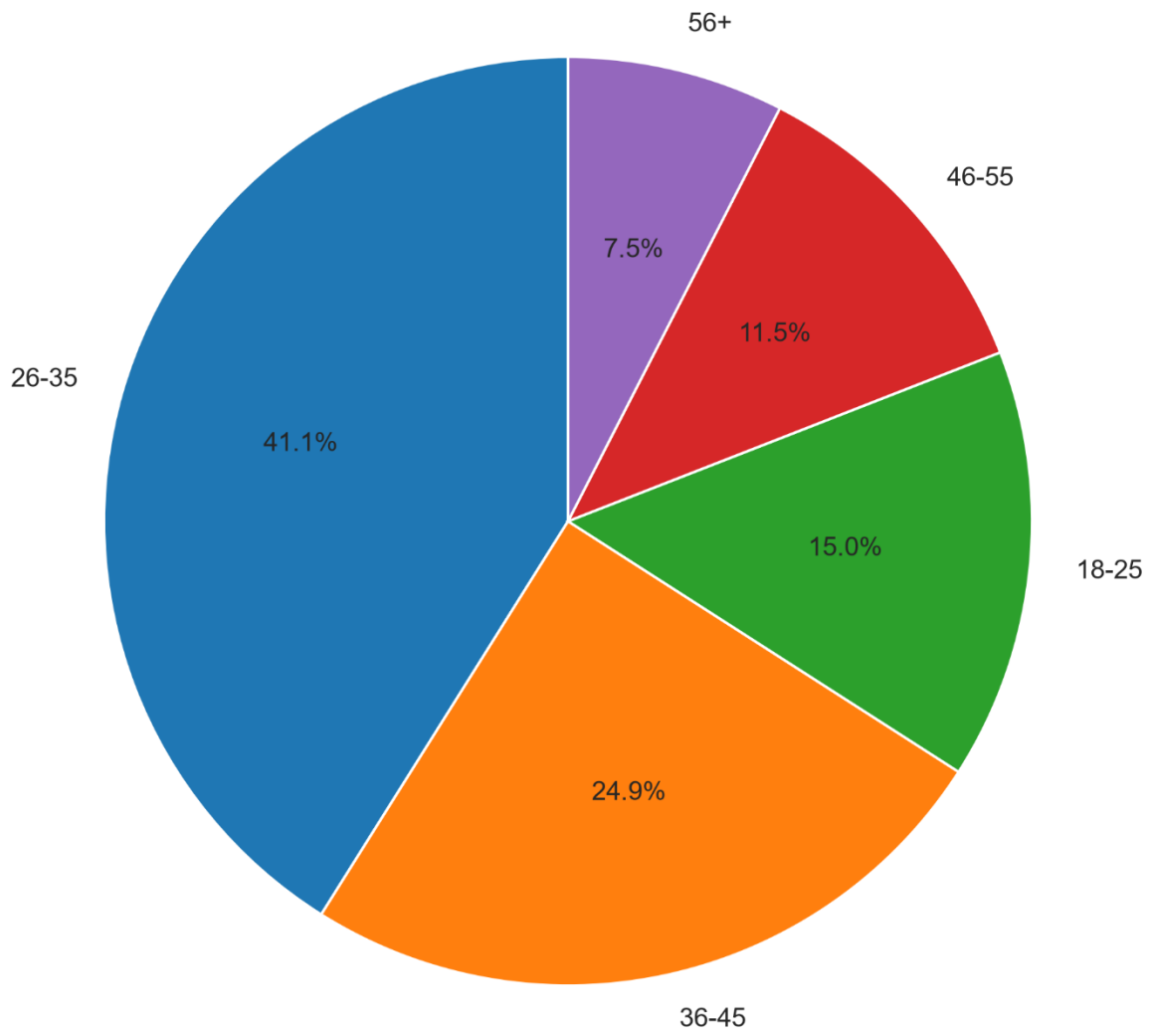
- Equity-oriented categories consistently attracted positive net inflows.
- Certain debt categories experienced fluctuating investor demand.
- Seasonal variations in investment behavior were observed.

9.6 Investor Demographics

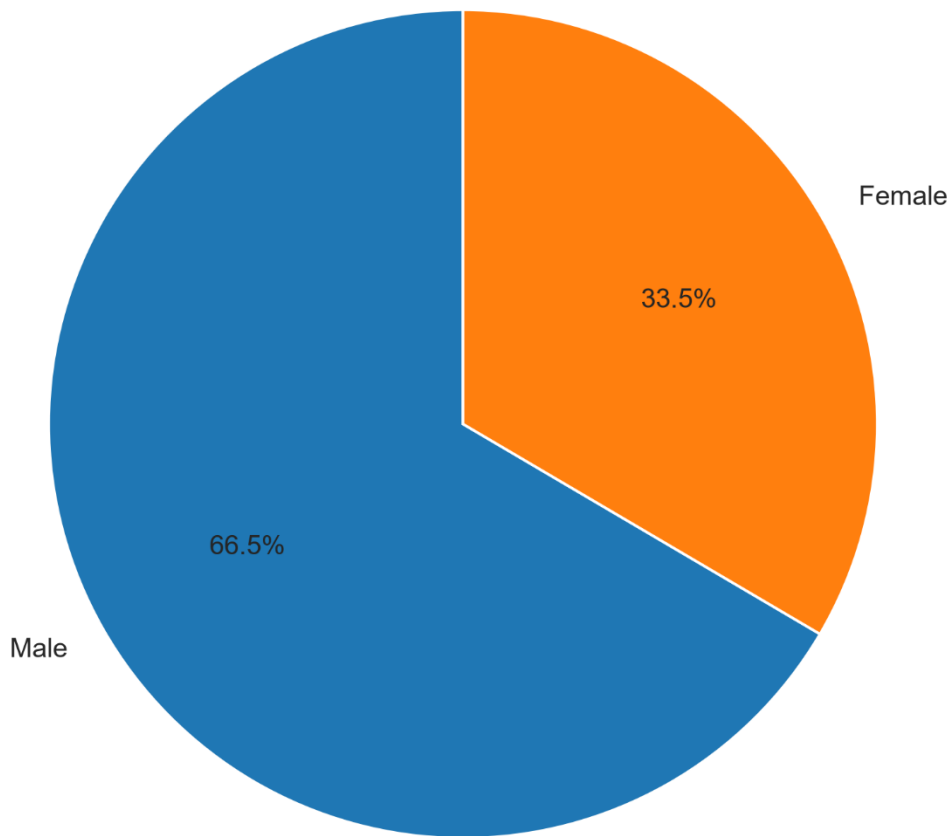
Investor demographic data was analyzed based on age group, gender, and investment characteristics.



Investor Distribution by Age Group



Gender Distribution of Investors

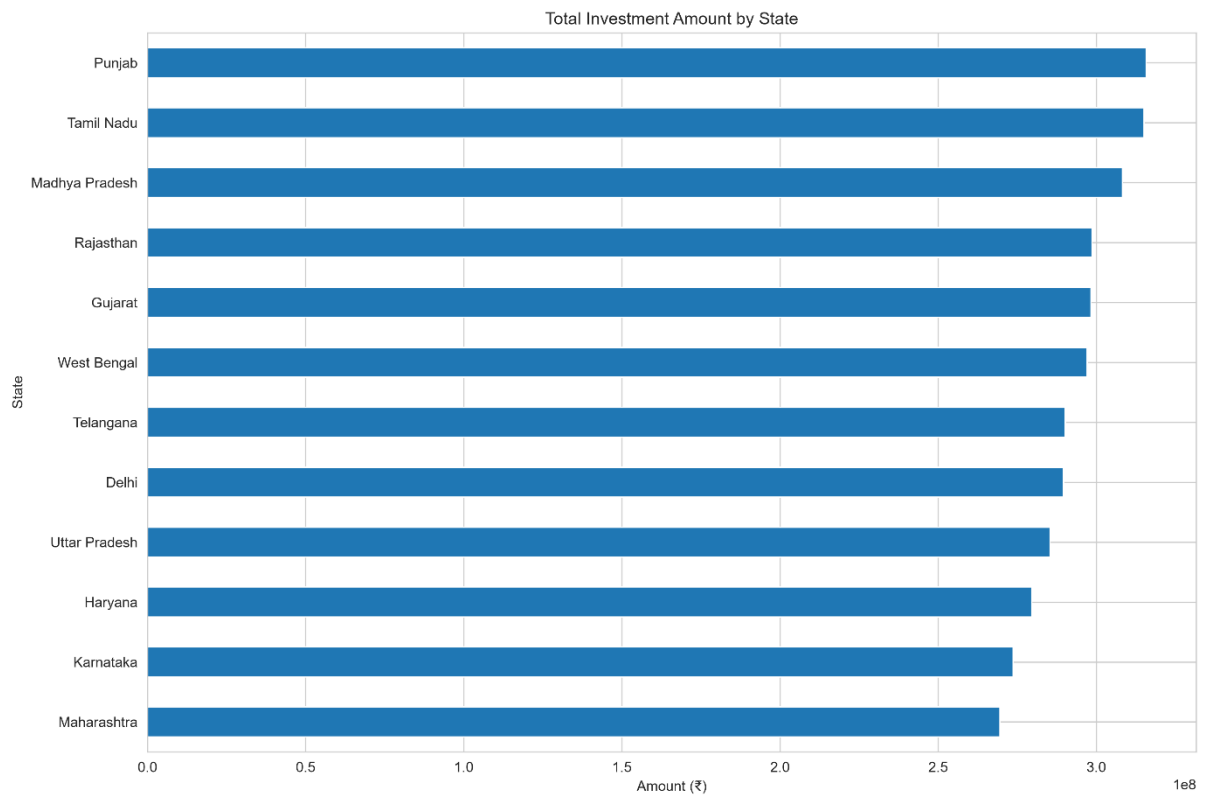


Observation

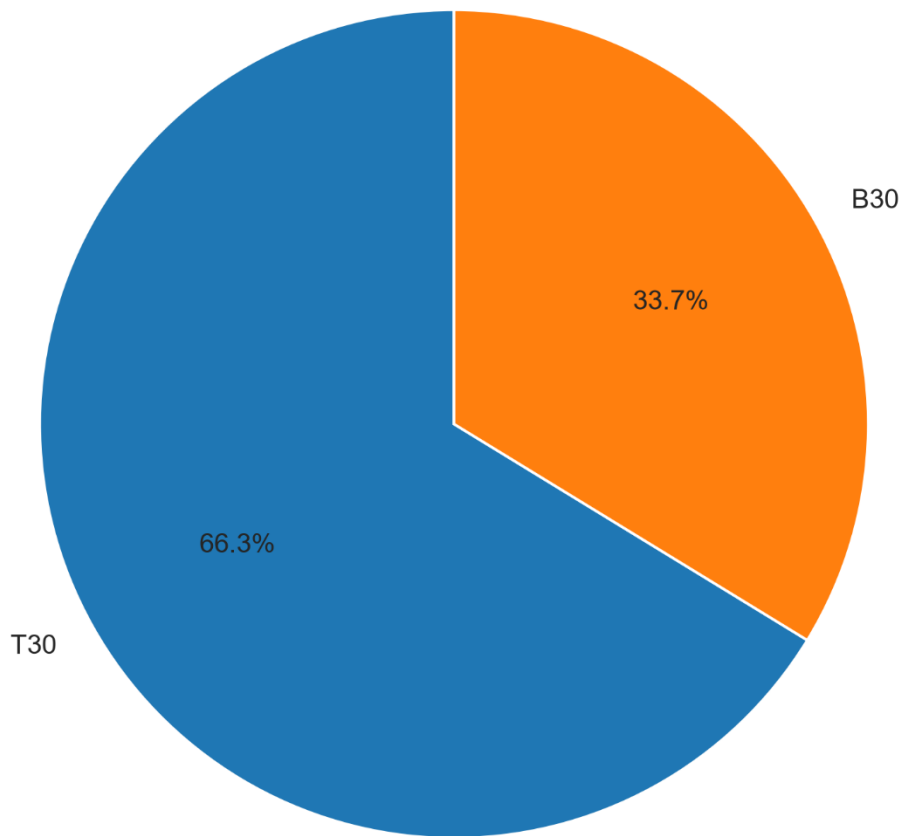
- Middle-aged investors represented the largest share of SIP participants.
- Male and female participation showed measurable differences in investment volume.
- Average SIP amounts varied across age groups.

9.7 Geographic Distribution

Investment activity was evaluated across different states and city classifications.



T30 vs B30 Investor Distribution

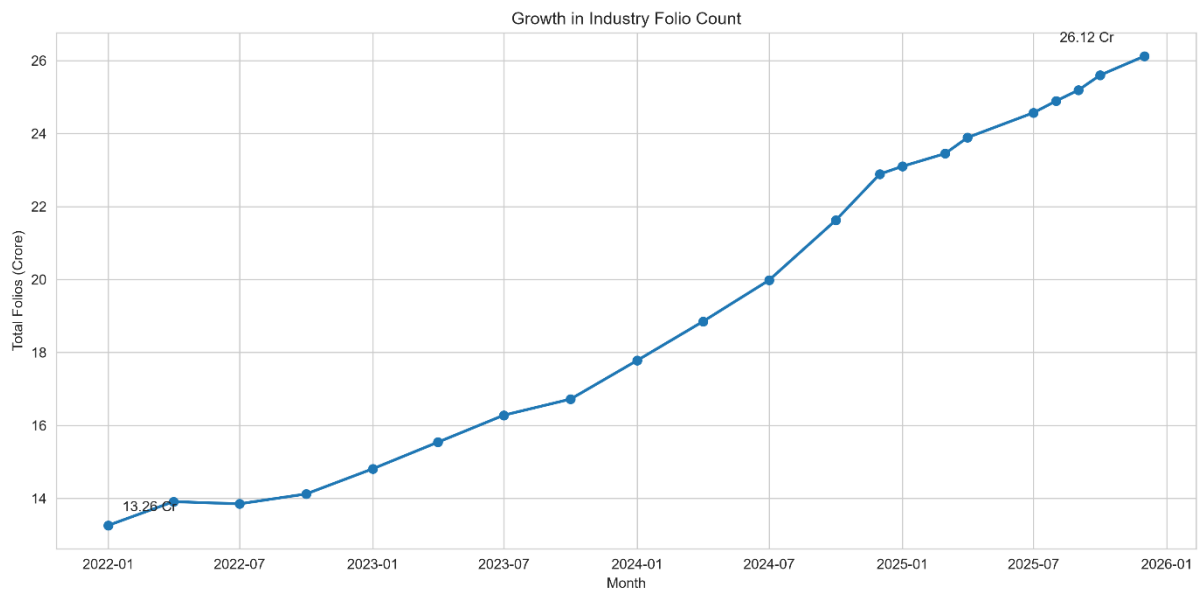


Observation

- Metropolitan regions contributed a significant proportion of total investments.
- Emerging cities also demonstrated increasing participation through SIP investments.
- Geographic diversification suggests expanding mutual fund adoption.

9.8 Industry Folio Growth

The number of mutual fund folios was analyzed over time to measure investor growth.

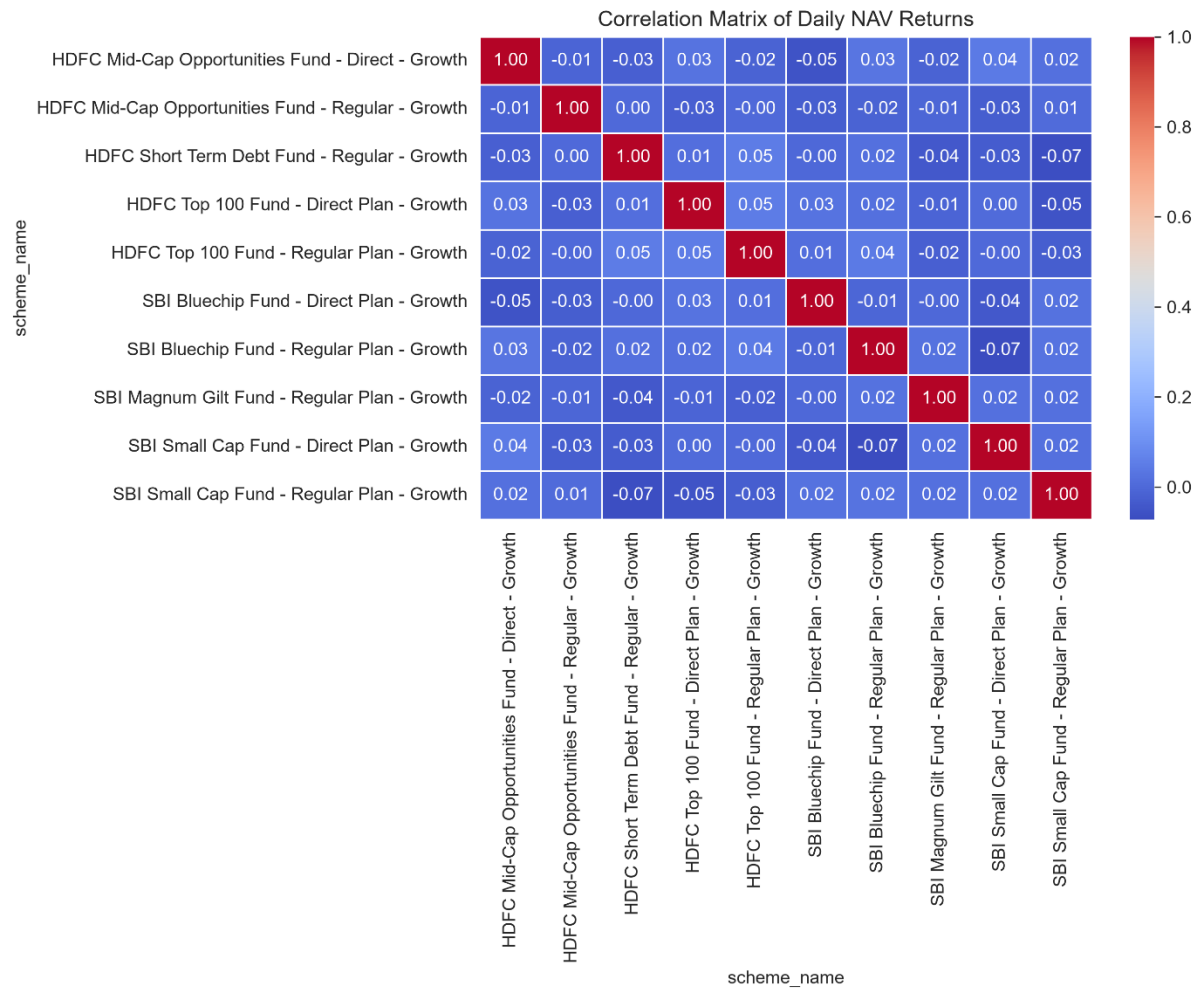


Observation

- Folio counts increased consistently throughout the study period.
- The trend reflects sustained expansion of retail investor participation.

9.9 Correlation Analysis

Daily returns for selected mutual fund schemes were compared using a correlation matrix.



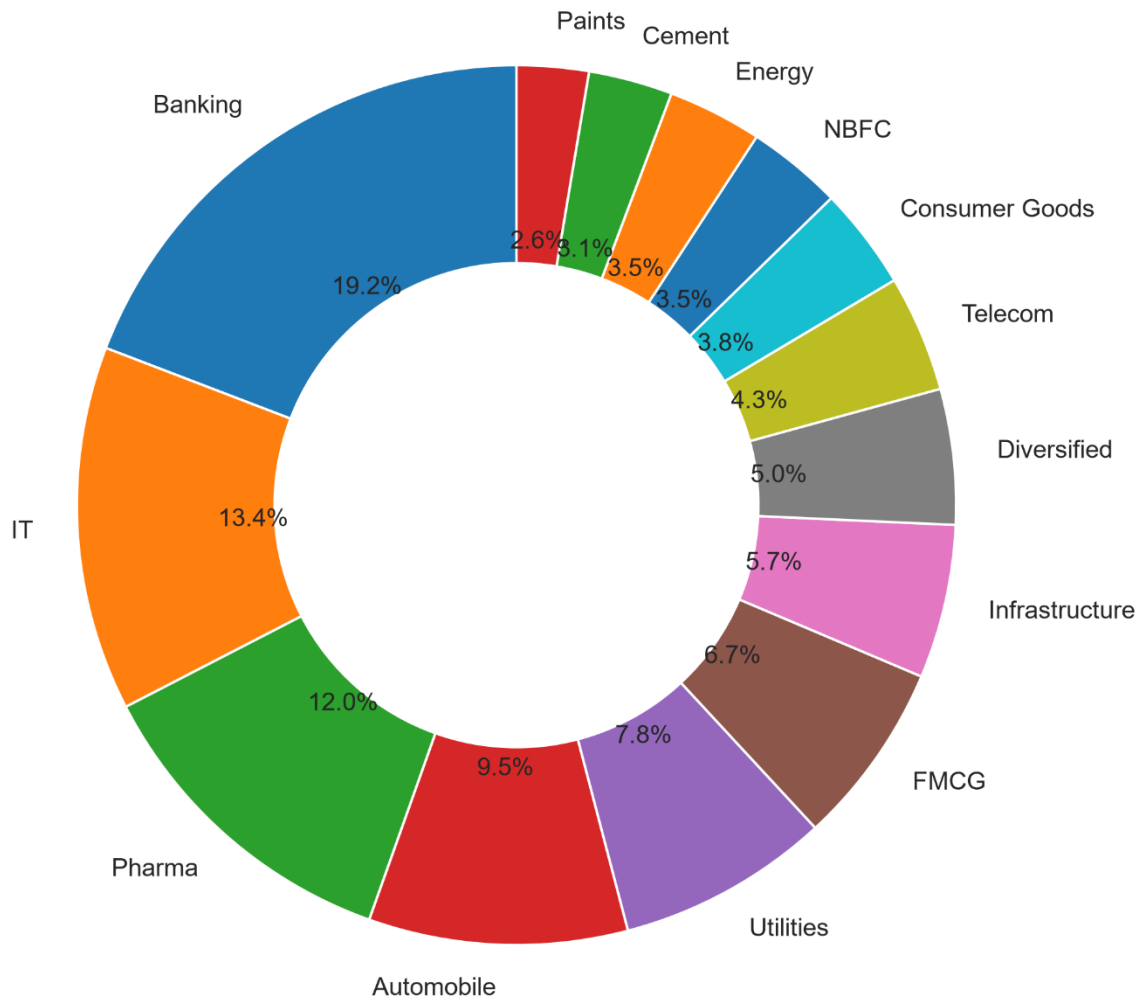
Observation

- Several funds exhibited strong positive correlation due to similar investment styles.
- Moderate diversification opportunities exist across categories.

9.10 Portfolio Sector Allocation

Portfolio holdings were aggregated to analyze sector exposure across equity funds.

Sector Allocation Across Equity Funds



Observation

- Financial Services and Information Technology represented major portfolio allocations.
- Sector diversification differed significantly between schemes.

9.11 Summary of EDA Findings

The exploratory analysis revealed steady industry growth, increasing SIP participation, expanding investor base, and strong long-term performance across equity-oriented mutual funds. The results provided the foundation for the

quantitative performance metrics and advanced risk analyses presented in subsequent sections.

10. Performance Analytics

10.1 Overview

To evaluate mutual fund performance beyond absolute returns, multiple risk-adjusted financial metrics were calculated. These metrics measure return generation, downside risk, benchmark sensitivity, and portfolio resilience.

The analysis included CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Fund Scorecard, and Benchmark Comparison.

10.2 CAGR Analysis

Compound Annual Growth Rate (CAGR) was calculated for each mutual fund scheme to measure annualized growth over multiple investment horizons.

	A	B	C	D	E	F
1	amfi_code	CAGR_1Y	CAGR_3Y	CAGR_5Y	scheme_name	
2	amfi_code	-2.224271	1.2926493	2.3168435	HDFC Top 100 Fund - Regular Plan - Growth	
3	amfi_code	3.7049686	3.91639	3.9126526	HDFC Short Term Debt Fund - Regular - Growth	
4	amfi_code	53.232396	32.442459	26.074068	HDFC Mid-Cap Opportunities Fund - Regular - Growth	
5	amfi_code	47.92412	28.967635	20.44273	ABSL Frontline Equity Fund - Regular - Growth	
6	amfi_code	-23.98603	-4.152381	6.9533357	ABSL Small Cap Fund - Regular - Growth	
7	amfi_code	7.2366453	6.3157845	5.7057342	ABSL Liquid Fund - Regular - Growth	
8	amfi_code	20.207704	19.667262	15.878823	UTI Nifty 50 Index Fund - Regular - Growth	
9	amfi_code	-16.79748	-0.767406	1.0303498	UTI Mid Cap Fund - Regular - Growth	
10	amfi_code	13.583135	25.556188	14.67811	UTI Flexi Cap Fund - Regular - Growth	
11	amfi_code	33.981048	22.65236	20.88118	Nippon India Large Cap Fund - Regular - Growth	
12	amfi_code	25.53998	21.089977	14.188358	Nippon India Large Cap Fund - Direct - Growth	
13	amfi_code	15.811222	8.0079277	14.274127	Nippon India Small Cap Fund - Regular - Growth	
14	amfi_code	22.493225	20.008208	13.914387	Nippon India ETF Nifty 50 BeES	
15	amfi_code	10.45494	4.062205	4.663928	Nippon India Gilt Securities Fund - Regular - Growth	
16	amfi_code	1.6091187	0.5258871	5.582555	Axis Bluechip Fund - Regular - Growth	
17	amfi_code	19.766741	20.816771	6.917406	Axis Bluechip Fund - Direct - Growth	
18	amfi_code	22.261065	35.111802	24.445309	Axis Midcap Fund - Regular - Growth	
19	amfi_code	-42.79761	-11.70581	1.3376248	Axis Small Cap Fund - Regular - Growth	
20	amfi_code	5.5219955	5.8403588	5.1644357	SBI Magnum Gilt Fund - Regular Plan - Growth	
21	amfi_code	60.437341	30.456524	22.384873	SBI Bluechip Fund - Regular Plan - Growth	
22	amfi_code	5.7061	16.267372	18.713119	SBI Bluechip Fund - Direct Plan - Growth	
23	amfi_code	82.776059	26.669922	28.033224	SBI Small Cap Fund - Regular Plan - Growth	
24	amfi_code	13.947803	-1.337387	1.8056214	SBI Small Cap Fund - Direct Plan - Growth	
25	amfi_code	8.4890813	13.978382	16.355566	ICICI Pru Bluechip Fund - Regular - Growth	
26	amfi_code	13.064279	32.487429	20.234028	ICICI Pru Bluechip Fund - Direct - Growth	
27	amfi_code	29.604659	31.777537	28.376762	ICICI Pru Midcap Fund - Regular - Growth	
28	amfi_code	41.819048	18.421997	14.888597	ICICI Pru Value Discovery Fund - Regular - Growth	
29	amfi_code	7.4250508	7.3942863	6.3416903	ICICI Pru Liquid Fund - Regular - Growth	
30	amfi_code	16.837098	17.267275	11.815442	Kotak Bluechip Fund - Regular - Growth	
31	amfi_code	22.501957	7.7560036	5.8738424	Kotak Emerging Equity Fund - Regular - Growth	
32	amfi_code	26.657082	29.58277	26.742494	Kotak Flexicap Fund - Regular - Growth	
33	amfi_code	7.120323	6.6959716	6.0604314	Kotak Liquid Fund - Regular - Growth	
34	amfi_code	28.675358	18.434489	16.555919	HDFC Top 100 Fund - Direct Plan - Growth	
35	amfi_code	15.443721	11.245223	9.8801991	HDFC Mid-Cap Opportunities Fund - Direct - Growth	
36	amfi_code	20.360678	34.000916	26.799277	Mirae Asset Large Cap Fund - Regular - Growth	
37	amfi_code	14.269394	14.748036	19.56596	Mirae Asset Emerging Bluechip Fund - Regular - Growth	
38	amfi_code	39.751761	29.178902	27.62986	Mirae Asset Tax Saver Fund - Regular - Growth	
39	amfi_code	18.11153	18.044206	11.695027	DSP Top 100 Equity Fund - Regular - Growth	
40	amfi_code	21.481222	26.869991	25.611807	DSP Midcap Fund - Regular - Growth	
41	amfi_code	65.138719	27.000427	27.91743	DSP Small Cap Fund - Regular - Growth	

Observation

- Several schemes delivered strong long-term annualized returns.
- Growth rates varied across categories.

10.3 Sharpe Ratio

The Sharpe Ratio measured excess return per unit of total risk.

	A	B	C	D
1	amfi_code	Sharpe_Ratio	scheme_name	
2	148567	1.4482912	Mirae Asset Large Cap Fund - Regular - Growth	
3	120843	1.306744	Kotak Flexicap Fund - Regular - Growth	
4	148569	1.23493	Mirae Asset Tax Saver Fund - Regular - Growth	
5	119551	1.2082666	SBI Bluechip Fund - Regular Plan - Growth	
6	120505	1.1801012	ICICI Pru Midcap Fund - Regular - Growth	
7	149323	1.1321219	DSP Midcap Fund - Regular - Growth	
8	100033	1.0936987	HDFC Mid-Cap Opportunities Fund - Regular - Growth	
9	118632	1.0816532	Nippon India Large Cap Fund - Regular - Growth	
10	101206	1.0272133	ABSL Frontline Equity Fund - Regular - Growth	
11	120504	1.0265243	ICICI Pru Bluechip Fund - Direct - Growth	
12	119094	0.998231	Axis Midcap Fund - Regular - Growth	
13	119552	0.9532795	SBI Bluechip Fund - Direct Plan - Growth	
14	149324	0.9437958	DSP Small Cap Fund - Regular - Growth	
15	119598	0.9453083	SBI Small Cap Fund - Regular Plan - Growth	
16	148568	0.9262873	Mirae Asset Emerging Bluechip Fund - Regular - Growth	
17	102885	0.8170993	UTI Nifty 50 Index Fund - Regular - Growth	
18	120503	0.7979729	ICICI Pru Bluechip Fund - Regular - Growth	
19	125497	0.7757902	HDFC Top 100 Fund - Direct Plan - Growth	
20	118635	0.664857	Nippon India ETF Nifty 50 BeES	
21	120506	0.6488788	ICICI Pru Value Discovery Fund - Regular - Growth	
22	118633	0.6452074	Nippon India Large Cap Fund - Direct - Growth	
23	102887	0.6195176	UTI Flexi Cap Fund - Regular - Growth	
24	120507	0.4957234	ICICI Pru Liquid Fund - Regular - Growth	
25	120841	0.4942725	Kotak Bluechip Fund - Regular - Growth	
26	149322	0.4640374	DSP Top 100 Equity Fund - Regular - Growth	
27	118634	0.448434	Nippon India Small Cap Fund - Regular - Growth	
28	125498	0.3006119	HDFC Mid-Cap Opportunities Fund - Direct - Growth	
29	101207	0.1626608	ABSL Small Cap Fund - Regular - Growth	
30	119093	0.1296145	Axis Bluechip Fund - Direct - Growth	
31	120842	0.0759998	Kotak Emerging Equity Fund - Regular - Growth	
32	119092	0.0307849	Axis Bluechip Fund - Regular - Growth	
33	119599	-0.057187	SBI Small Cap Fund - Direct Plan - Growth	
34	119095	-0.075974	Axis Small Cap Fund - Regular - Growth	
35	120844	-0.088747	Kotak Liquid Fund - Regular - Growth	
36	100016	-0.201517	HDFC Top 100 Fund - Regular Plan - Growth	
37	102886	-0.205832	UTI Mid Cap Fund - Regular - Growth	
38	119120	-0.226575	SBI Magnum Gilt Fund - Regular Plan - Growth	
39	118636	-0.35663	Nippon India Gilt Securities Fund - Regular - Growth	
40	100025	-0.567095	HDFC Short Term Debt Fund - Regular - Growth	
41	101208	-0.815567	ABSL Liquid Fund - Regular - Growth	

Observation

- Higher Sharpe Ratios indicated superior risk-adjusted performance.
- Top-ranked funds consistently balanced return generation with controlled volatility.

10.4 Sortino Ratio

The Sortino Ratio focused exclusively on downside risk by penalizing only negative return volatility.

Observation

- Several schemes exhibited stronger downside protection than indicated by the Sharpe Ratio alone.

10.5 Alpha and Beta

Regression analysis against benchmark indices was performed to calculate Alpha and Beta.

	A	B	C	D
1	amfi_code	Alpha	Beta	scheme_name
2	100016	0.037475808	-0.058268431	HDFC Top 100 Fund - Regular Plan - Growth
3	100025	0.04281792	0.001158218	HDFC Short Term Debt Fund - Regular - Growth
4	100033	0.271953545	0.005103526	HDFC Mid-Cap Opportunities Fund - Regular - Growth
5	101206	0.213997848	0.021086316	ABSL Frontline Equity Fund - Regular - Growth
6	101207	0.108970919	-0.065289248	ABSL Small Cap Fund - Regular - Growth
7	101208	0.060861458	0.00026671	ABSL Liquid Fund - Regular - Growth
8	102885	0.170487568	-0.019486786	UTI Nifty 50 Index Fund - Regular - Growth
9	102886	0.028969398	-0.042125029	UTI Mid Cap Fund - Regular - Growth
10	102887	0.162112568	0.016682689	UTI Flexi Cap Fund - Regular - Growth
11	118632	0.218233979	-0.008353832	Nippon India Large Cap Fund - Regular - Growth
12	118633	0.156438359	-0.036639007	Nippon India Large Cap Fund - Direct - Growth
13	118634	0.175007217	0.103496637	Nippon India Small Cap Fund - Regular - Growth
14	118635	0.151363546	-0.001433521	Nippon India ETF Nifty 50 BeES
15	118636	0.05074773	0.001256674	Nippon India Gilt Securities Fund - Regular - Growth
16	119092	0.068995298	0.009730745	Axis Bluechip Fund - Regular - Growth
17	119093	0.082328013	0.025883212	Axis Bluechip Fund - Direct - Growth
18	119094	0.260766631	-0.066265429	Axis Midcap Fund - Regular - Growth
19	119095	0.048015954	-0.066951175	Axis Small Cap Fund - Regular - Growth
20	119120	0.056208952	-0.006413851	SBI Magnum Gilt Fund - Regular Plan - Growth
21	119551	0.232010072	-0.031751441	SBI Bluechip Fund - Regular Plan - Growth
22	119552	0.198685994	-0.026159308	SBI Bluechip Fund - Direct Plan - Growth
23	119598	0.303369654	-0.023196062	SBI Small Cap Fund - Regular Plan - Growth
24	119599	0.048824216	0.062001804	SBI Small Cap Fund - Direct Plan - Growth
25	120503	0.177033024	-0.040268952	ICICI Pru Bluechip Fund - Regular - Growth
26	120504	0.21194827	0.01623217	ICICI Pru Bluechip Fund - Direct - Growth
27	120505	0.292635833	0.000549004	ICICI Pru Midcap Fund - Regular - Growth
28	120506	0.162538533	0.041896463	ICICI Pru Value Discovery Fund - Regular - Growth
29	120507	0.067462104	-0.000444093	ICICI Pru Liquid Fund - Regular - Growth
30	120841	0.130429245	0.036356399	Kotak Bluechip Fund - Regular - Growth
31	120842	0.078044389	0.018056874	Kotak Emerging Equity Fund - Regular - Growth
32	120843	0.27330465	-0.02282957	Kotak Flexicap Fund - Regular - Growth
33	120844	0.064556775	-0.000400744	Kotak Liquid Fund - Regular - Growth
34	125497	0.177297549	0.048820404	HDFC Top 100 Fund - Direct Plan - Growth
35	125498	0.11989101	0.009054413	HDFC Mid-Cap Opportunities Fund - Direct - Growth
36	148567	0.269837514	0.023683558	Mirae Asset Large Cap Fund - Regular - Growth
37	148568	0.208730101	-0.02638772	Mirae Asset Emerging Bluechip Fund - Regular - Growth
38	148569	0.282703679	0.018133884	Mirae Asset Tax Saver Fund - Regular - Growth
39	149322	0.131731902	-0.004050271	DSP Top 100 Equity Fund - Regular - Growth
40	149323	0.265985775	-0.002523005	DSP Midcap Fund - Regular - Growth
41	149324	0.300578782	0.01145516	DSP Small Cap Fund - Regular - Growth

Observation

- Positive Alpha indicated fund managers generated returns beyond benchmark expectations.
- Beta values reflected varying levels of market sensitivity.

10.6 Maximum Drawdown

Maximum Drawdown measured the largest peak-to-trough decline experienced by each scheme.

Observation

- Equity-oriented funds experienced larger drawdowns during market corrections.
- Conservative funds demonstrated comparatively lower downside exposure.

10.7 Fund Scorecard

A composite scoring model was developed using multiple weighted performance indicators.

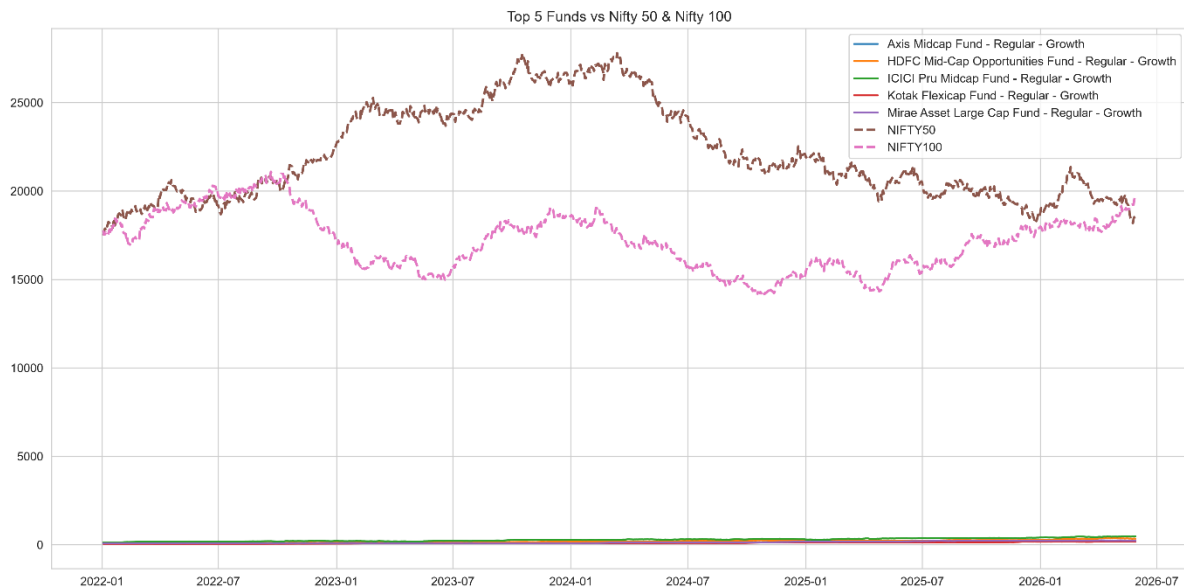
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	amfi_code	CAGR_3Y	Sharpe_Ratio	Alpha	expense_ratio	max_drawdown	scheme_name	return_ran	sharpe_ran	alpha_ran	expense_ran	drawdown_ran	Fund_Score
2	120505	31.777537	1.180102	0.2326358	1.36	-18.1885136	ICICI Pru Midcap Fund - Regular - Growth	0.9	0.3	0.35	0.65	0.625	84.5
3	119094	35.118802	0.936231	0.2607667	1.38	-20.9608839	Axis Midcap Fund - Regular - Growth	1	0.75	0.8	0.6	0.7	80.75
4	100033	32.442459	1.0936987	0.2719535	1.38	-16.2172086	HDFC Mid-Cap Opportunities Fund - Regular - Gr	0.925	0.85	0.875	0.6	0.5	80.5
5	148567	34.000916	1.4482312	0.2698375	1.46	-11.2657291	Mirae Asset Large Cap Fund - Regular - Growth	0.975	1	0.85	0.45	0.2	80
6	120843	29.58277	1.306744	0.2733047	1.45	-12.3739682	Kotak Flexicap Fund - Regular - Growth	0.85	0.975	0.9	0.475	0.325	78.25
7	120504	32.487429	1.0265243	0.2119483	0.8	-12.5882756	ICICI Pru Bluechip Fund - Direct - Growth	0.95	0.775	0.7	0.725	0.3	75.75
8	119598	26.669322	0.9453083	0.3033637	1.43	-28.7060057	SBI Small Cap Fund - Regular Plan - Growth	0.725	0.675	1	0.5	0.9	75.125
9	149324	27.000427	0.9497958	0.3005788	1.52	-31.1719003	DSP Small Cap Fund - Regular - Growth	0.775	0.7	0.975	0.3625	0.925	74.9375
10	148569	29.178902	1.23493	0.2827037	1.6	-16.3967429	Mirae Asset Tax Saver Fund - Regular - Growth	0.825	0.95	0.925	0.1125	0.525	73.9375
11	119551	30.456524	1.2082666	0.2320101	1.54	-15.0123855	SBI Bluechip Fund - Regular Plan - Growth	0.875	0.925	0.775	0.2625	0.425	73.0625
12	118632	22.65236	1.0816592	0.2182394	1.51	-17.4140751	Nippon India Large Cap Fund - Regular - Growth	0.675	0.825	0.75	0.4125	0.575	67.6125
13	149323	26.869991	1.1321219	0.2659858	1.61	-17.2481058	DSP Midcap Fund - Regular - Growth	0.75	0.875	0.825	0.075	0.55	67.5
14	101206	28.967695	1.0272133	0.2139978	1.6	-11.2915363	ABSL Frontline Equity Fund - Regular - Growth	0.8	0.8	0.725	0.1125	0.225	62.4375
15	118633	21.089977	0.6452074	0.1564384	0.72	-18.6297431	Nippon India Large Cap Fund - Direct - Growth	0.65	0.5	0.475	0.8875	0.65	61.3125
16	119552	16.267372	0.9532795	0.188686	0.66	-11.8034935	SBI Bluechip Fund - Direct Plan - Growth	0.45	0.725	0.65	0.925	0.275	61.25
17	125497	18.434489	0.7757902	0.1772975	0.92	-15.2000131	HDFC Top 100 Fund - Direct Plan - Growth	0.55	0.575	0.625	0.675	0.45	58
18	120506	18.421997	0.6488788	0.1625385	1.41	-19.8256583	ICICI Pru Value Discovery Fund - Regular - Growth	0.525	0.525	0.525	0.55	0.675	54.375
19	118635	20.008208	0.664857	0.1513635	0.89	-11.6506245	Nippon India ETF Nifty 50 BeES	0.6	0.55	0.45	0.7	0.25	53.75
20	119093	20.816771	0.1296145	0.082328	0.75	-21.7513964	Axis Bluechip Fund - Direct - Growth	0.625	0.3	0.325	0.825	0.775	52.875
21	148568	14.748036	0.9262873	0.2087301	1.52	-15.2725942	Mirae Asset Emerging Bluechip Fund - Regular - C	0.425	0.65	0.675	0.3625	0.475	52.6875
22	102887	25.556168	0.6195176	0.1621126	1.64	-21.5398271	UTI Flexi Cap Fund - Regular - Growth	0.7	0.475	0.5	0.0375	0.75	50.9375
23	120503	13.978362	0.7979729	0.177033	1.42	-13.8978628	ICICI Pru Bluechip Fund - Regular - Growth	0.4	0.6	0.6	0.525	0.35	50.375
24	102885	19.667262	0.8170993	0.1704876	1.57	-10.8598622	UTI Nifty 50 Index Fund - Regular - Growth	0.575	0.625	0.55	0.175	0.175	48.25
25	125498	11.245223	0.3006119	0.119891	0.78	-21.1172796	HDFC Mid-Cap Opportunities Fund - Direct - Grow	0.375	0.35	0.375	0.775	0.725	46.375
26	118634	8.0079277	0.448434	0.1750072	1.53	-23.3448861	Nippon India Small Cap Fund - Regular - Growth	0.35	0.375	0.575	0.3125	0.8	44.0625
27	149322	18.044206	0.4640374	0.1317319	1.54	-14.8445769	DSP Top 100 Equity Fund - Regular - Growth	0.5	0.4	0.425	0.2625	0.4	41.4375
28	120841	17.267275	0.4942725	0.1304232	1.59	-17.5736142	Kotak Bluechip Fund - Regular - Growth	0.475	0.425	0.4	0.15	0.6	41.125
29	120507	7.3942863	0.4957234	0.0674621	0.74	-0.0977314	ICICI Pru Liquid Fund - Regular - Growth	0.3	0.45	0.25	0.85	0.025	38.25
30	120842	7.7560036	0.0759998	0.0780444	1.56	-24.0035113	Kotak Emerging Equity Fund - Regular - Growth	0.325	0.275	0.3	0.2	0.825	33.875
31	119599	-1.337387	-0.057187	0.0488242	0.72	-52.5742212	SBI Small Cap Fund - Direct Plan - Growth	0.075	0.225	0.125	0.8875	1	33.6875
32	120844	6.6959716	-0.088747	0.0645568	0.6	-0.11629328	Kotak Liquid Fund - Regular - Growth	0.275	0.175	0.225	0.95	0.05	31.875
33	101207	-4.152381	0.1626608	0.1089709	1.53	-35.4469165	ABSL Small Cap Fund - Regular - Growth	0.05	0.325	0.35	0.3125	0.95	30.8125
34	118636	4.062205	-0.35663	0.0507477	0.55	-8.31636176	Nippon India Gilt Securities Fund - Regular - Grow	0.2	0.075	0.15	1	0.15	27.375
35	119095	-11.70581	-0.0759374	0.048016	1.38	-51.6777537	Axis Small Cap Fund - Regular - Growth	0.025	0.2	0.1	0.6	0.975	26.5
36	119120	5.8403588	-0.226575	0.056209	0.77	-4.32872892	SBI Magnum Gilt Fund - Regular Plan - Growth	0.225	0.1	0.175	0.8	0.125	26
37	101208	6.3157845	-0.815567	0.0608615	0.79	-0.1622498	ABSL Liquid Fund - Regular - Growth	0.25	0.025	0.2	0.75	0.075	24.125
38	100025	3.91639	-0.567095	0.0428179	0.56	-4.30826382	HDFC Short Term Debt Fund - Regular - Growth	0.175	0.05	0.075	0.975	0.1	23.625
39	102886	-0.767406	-0.205832	0.0289694	1.51	-28.0011236	UTI Mid Cap Fund - Regular - Growth	0.1	0.125	0.025	0.4125	0.875	21.5625
40	100016	1.2926493	-0.201517	0.0374758	1.55	-24.7344409	HDFC Top 100 Fund - Regular Plan - Growth	0.15	0.15	0.05	0.225	0.85	21.125
41	119092	0.5258871	0.0307849	0.0689953	1.64	-14.4016179	Axis Bluechip Fund - Regular - Growth	0.125	0.25	0.275	0.0375	0.375	19.8125

Observation

The scorecard enabled comprehensive comparison of all mutual funds using multiple performance dimensions rather than a single metric.

10.8 Benchmark Comparison

Fund performance was compared against benchmark indices.



Observation

- Several schemes consistently outperformed benchmark indices over the evaluation period.
- Tracking error analysis highlighted differences in benchmark replication.

11. Advanced Analytics

11.1 Historical VaR & CvaR

Historical Value at Risk (VaR) and Conditional Value at Risk (CvaR) were computed to estimate downside risk using historical daily return distributions.

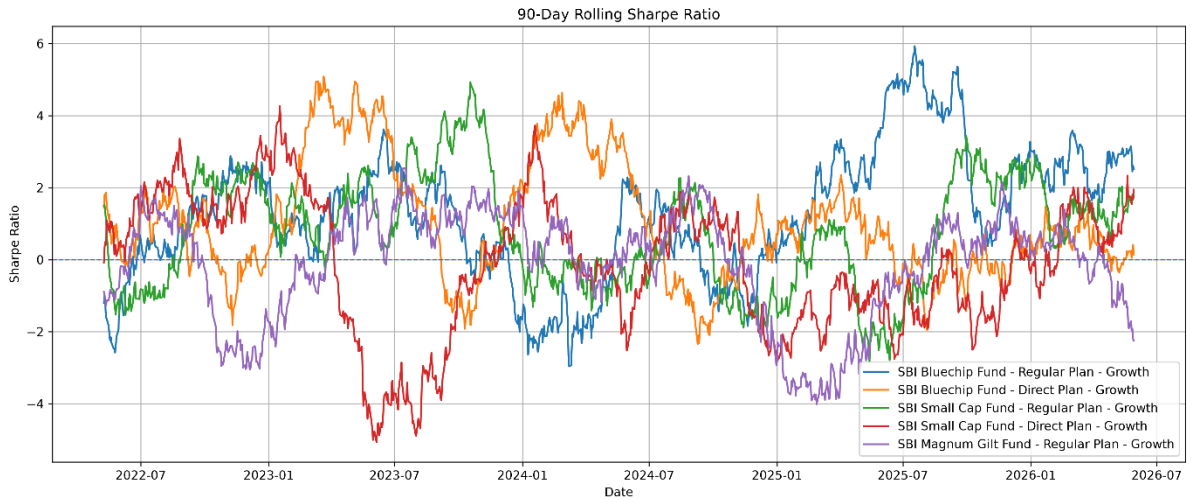
	A	B	C	D	E
1	amfi_Loading	VaR_95	CVaR_95	scheme_name	
2	119539	-0.026853	-0.032384	SBI Small Cap Fund - Direct Plan - Growth	
3	119095	-0.026188	-0.031667	Axis Small Cap Fund - Regular - Growth	
4	101207	-0.026021	-0.032459	ABSL Small Cap Fund - Regular - Growth	
5	118634	-0.025438	-0.032304	Nippon India Small Cap Fund - Regular - Growth	
6	119598	-0.024507	-0.030595	SBI Small Cap Fund - Regular Plan - Growth	
7	149324	-0.023483	-0.031036	DSP Small Cap Fund - Regular - Growth	
8	102886	-0.01922	-0.023251	UTI Mid Cap Fund - Regular - Growth	
9	100033	-0.019034	-0.023456	HDFC Mid-Cap Opportunities Fund - Regular - Growth	
10	120505	-0.018892	-0.024342	ICICI Pru Midcap Fund - Regular - Growth	
11	119094	-0.01848	-0.02426	Axis Midcap Fund - Regular - Growth	
12	125498	-0.018365	-0.023094	HDFC Mid-Cap Opportunities Fund - Direct - Growth	
13	120842	-0.018219	-0.022725	Kotak Emerging Equity Fund - Regular - Growth	
14	149323	-0.017882	-0.022607	DSP Midcap Fund - Regular - Growth	
15	148569	-0.017118	-0.021328	Mirae Asset Tax Saver Fund - Regular - Growth	
16	102887	-0.015232	-0.019411	UTI Flexi Cap Fund - Regular - Growth	
17	148568	-0.014712	-0.018765	Mirae Asset Emerging Bluechip Fund - Regular - Growth	
18	125497	-0.014532	-0.01831	HDFC Top 100 Fund - Direct Plan - Growth	
19	120843	-0.014508	-0.018381	Kotak Flexicap Fund - Regular - Growth	
20	149322	-0.014474	-0.018402	DSP Top 100 Equity Fund - Regular - Growth	
21	120506	-0.014388	-0.019405	ICICI Pru Value Discovery Fund - Regular - Growth	
22	100016	-0.014364	-0.01806	HDFC Top 100 Fund - Regular Plan - Growth	
23	119093	-0.014226	-0.017487	Axis Bluechip Fund - Direct - Growth	
24	118632	-0.013954	-0.017619	Nippon India Large Cap Fund - Regular - Growth	
25	119092	-0.01375	-0.017328	Axis Bluechip Fund - Regular - Growth	
26	120504	-0.013728	-0.017494	ICICI Pru Bluechip Fund - Direct - Growth	
27	148567	-0.01356	-0.017623	Mirae Asset Large Cap Fund - Regular - Growth	
28	120503	-0.01355	-0.017451	ICICI Pru Bluechip Fund - Regular - Growth	
29	119552	-0.013501	-0.017336	SBI Bluechip Fund - Direct Plan - Growth	
30	118633	-0.013443	-0.01663	Nippon India Large Cap Fund - Direct - Growth	
31	101206	-0.013282	-0.017439	ABSL Frontline Equity Fund - Regular - Growth	
32	120841	-0.013142	-0.016363	Kotak Bluechip Fund - Regular - Growth	
33	119551	-0.012846	-0.016397	SBI Bluechip Fund - Regular Plan - Growth	
34	102885	-0.012613	-0.01549	UTI Nifty 50 Index Fund - Regular - Growth	
35	118635	-0.012554	-0.016177	Nippon India ETF Nifty 50 BeES	
36	119120	-0.003938	-0.005014	SBI Magnum Gilt Fund - Regular Plan - Growth	
37	118636	-0.003804	-0.004919	Nippon India Gilt Securities Fund - Regular - Growth	
38	100025	-0.003793	-0.004994	HDFC Short Term Debt Fund - Regular - Growth	
39	120844	-0.000285	-0.000411	Kotak Liquid Fund - Regular - Growth	
40	101208	-0.000269	-0.000422	ABSL Liquid Fund - Regular - Growth	
41	120507	-0.000222	-0.000373	ICICI Pru Liquid Fund - Regular - Growth	

Observation

Funds with more negative VaR and CVaR values exhibited greater downside exposure during adverse market conditions.

11.2 Rolling 90-Day Sharpe Ratio

Rolling Sharpe Ratios were calculated to evaluate changes in risk-adjusted performance over time.

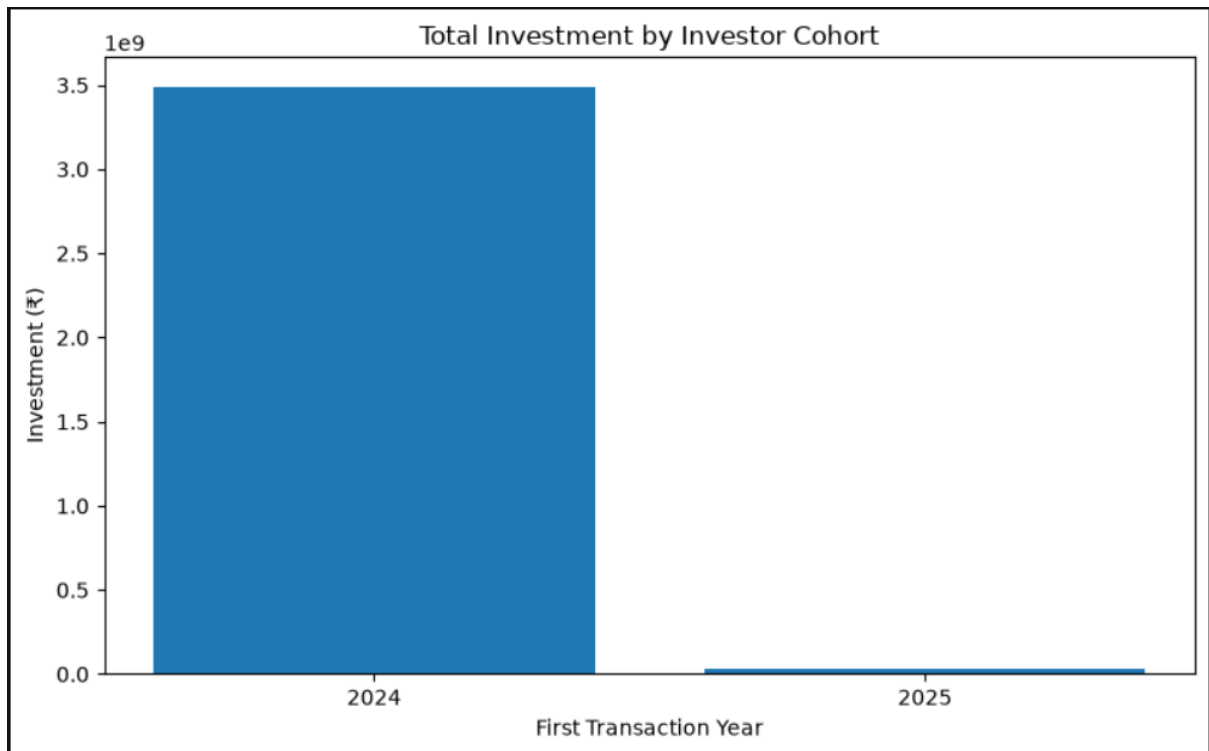


Observation

- Risk-adjusted performance fluctuated across market cycles.
- Temporary declines aligned with periods of increased market volatility.

11.3 Investor Cohort Analysis

Investors were grouped according to the year of their first investment.

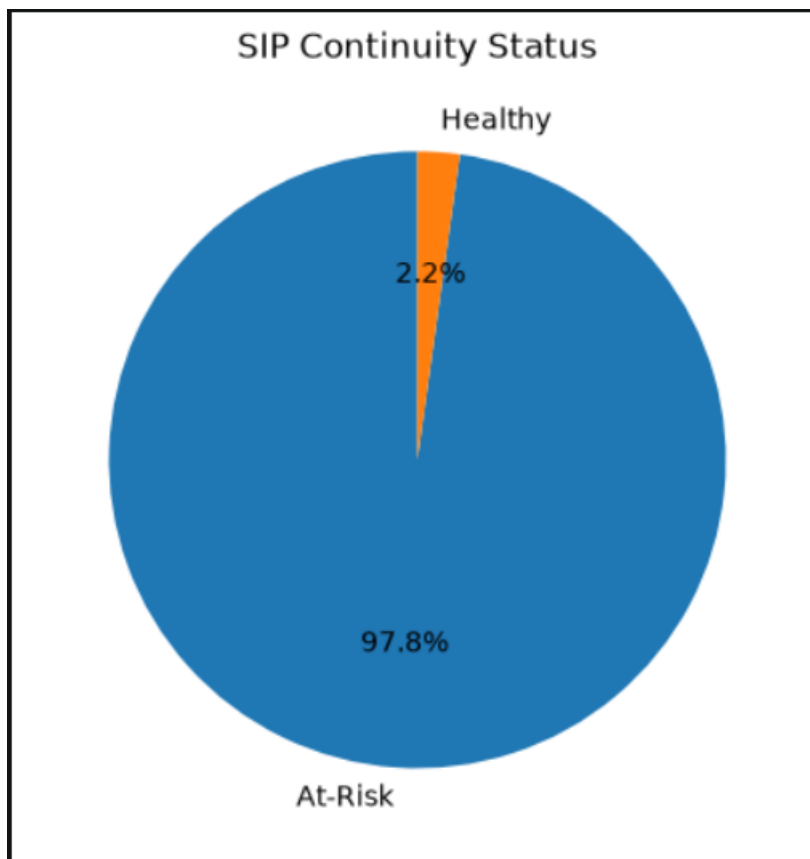


Observation

- Newer cohorts contributed significantly to investment growth.
- Average SIP amounts remained relatively consistent across cohorts.

11.4 SIP Continuity Analysis

Investment continuity was evaluated using transaction intervals.

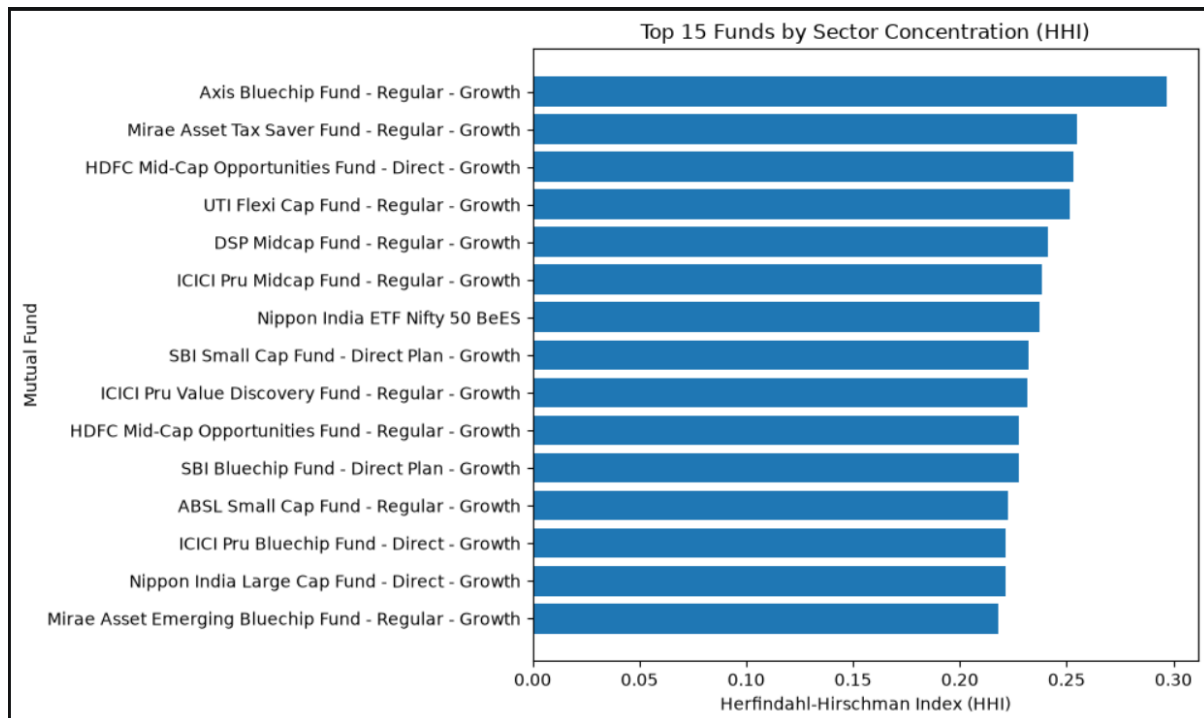


Observation

- Most investors maintained regular monthly SIP contributions.
- A subset of investors was identified as potentially at risk due to extended contribution gaps.

11.5 Sector Concentration (HHI)

The Herfindahl–Hirschman Index (HHI) was used to measure portfolio concentration.



Observation

- Lower HHI values indicated well-diversified portfolios.
- Higher HHI values reflected greater concentration within specific sectors.

11.6 Fund Recommendation System

A simple rule-based recommendation engine was developed using Sharpe Ratio rankings within predefined risk categories.

```
Available Risk Levels:
['High', 'Low', 'Moderate', 'Moderately High', 'Very High']

Enter Risk Appetite (Low/Moderate/High): Moderate

Top 3 Recommended Funds
```

scheme_name	fund_house	category	sharpe_ratio	return_3yr_pct	expense_ratio_pct
HDFC Top 100 Fund - Regular Plan - Growth	HDFC Mutual Fund	Large Cap	1.06	14.84	1.55
Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	Large Cap	1.06	14.81	1.46
ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	Large Cap	1.03	14.41	0.80

Observation

- Investors can obtain personalized recommendations based on their risk appetite.
- The recommendation logic combines risk classification with historical risk-adjusted performance.

12. Power BI Dashboard

12.1 Dashboard Overview

To enable interactive exploration of mutual fund performance and investor analytics, a four-page dashboard was developed using Microsoft Power BI. The dashboard integrates the cleaned datasets and calculated performance metrics into a unified business intelligence solution. Interactive features such as slicers, drill-through navigation, cross-filtering, KPI cards, and tooltips allow users to analyze mutual fund performance from multiple perspectives.

The dashboard was designed using the Bluestock color theme to provide a consistent and professional user experience.

12.2 Dashboard Pages

Page 1 – Industry Overview

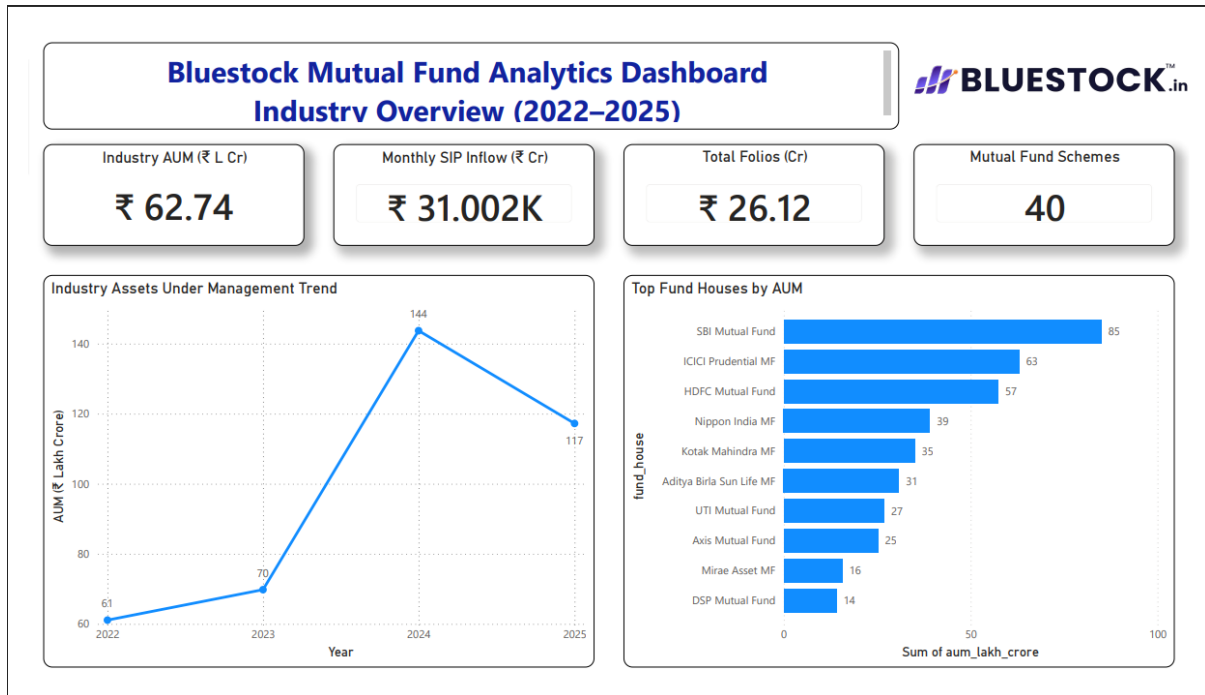
This page provides a high-level summary of the Indian mutual fund industry through key performance indicators and trend visualizations.

The dashboard includes:

- Total Assets Under Management (AUM)
- Total SIP Inflows
- Total Industry Folios
- Total Mutual Fund Schemes

- Industry AUM Trend
- AUM by Asset Management Company (AMC)

These visuals provide a quick overview of the industry's growth and market concentration.



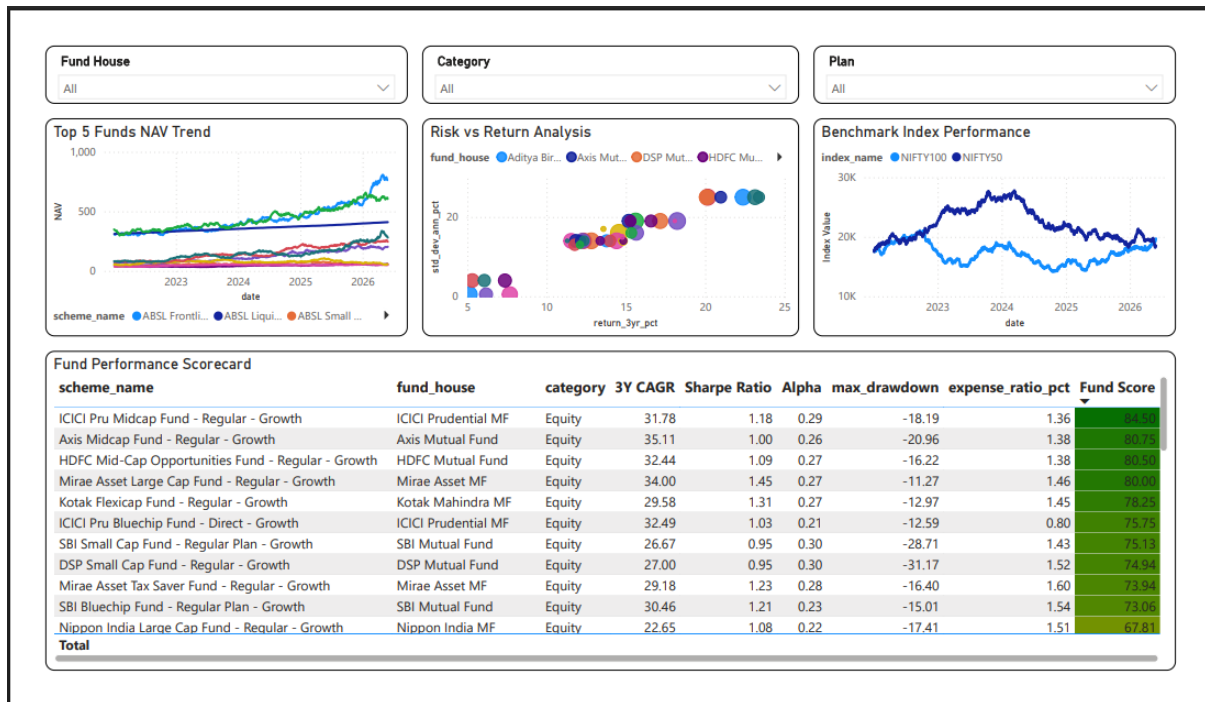
Page 2 – Fund Performance

This page focuses on evaluating mutual fund performance through multiple risk and return metrics.

The dashboard includes:

- Risk vs Return Scatter Plot
- Fund Scorecard Table
- NAV Trend compared with Benchmark
- Fund Performance Slicers
- Interactive Filtering

The scatter plot enables users to compare funds based on their expected returns and associated risk, while the scorecard consolidates multiple financial metrics for easier evaluation.



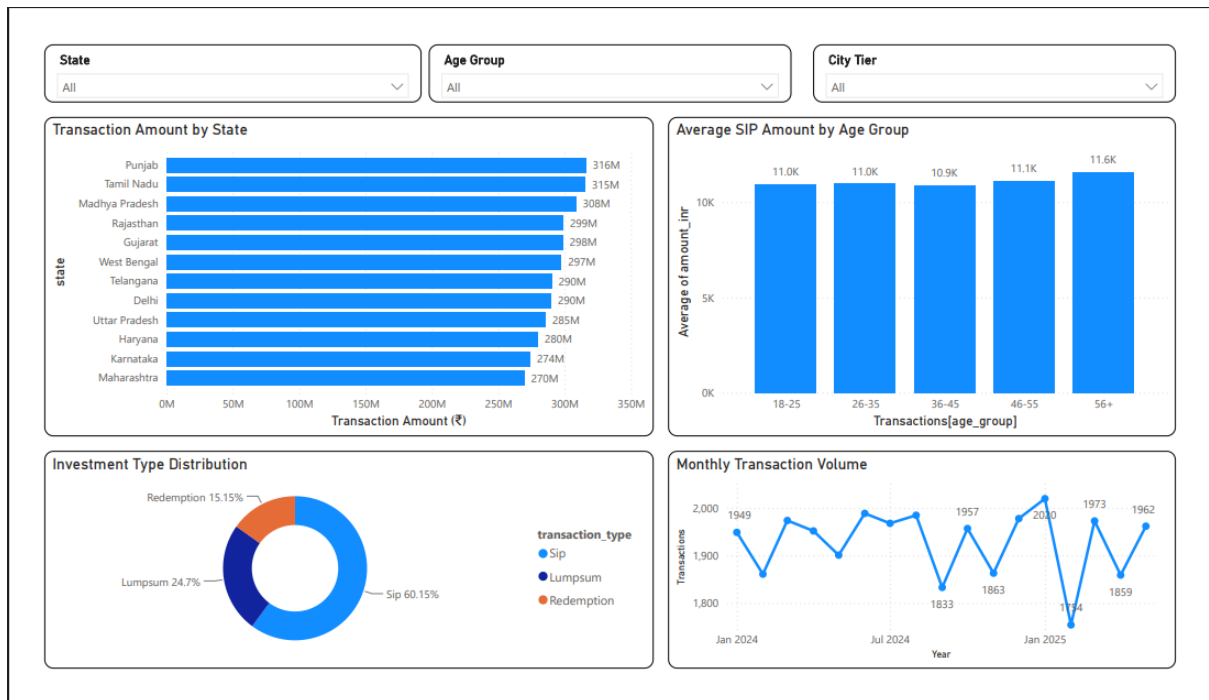
Page 3 – Investor Analytics

Investor behavior was analyzed using demographic and geographic information.

The dashboard includes:

- State-wise Transaction Amount
- Investment Type Distribution
- Age Group vs Average SIP Amount
- Monthly Transaction Volume
- Geographic Filters

These visualizations provide insights into investment patterns across different investor segments.



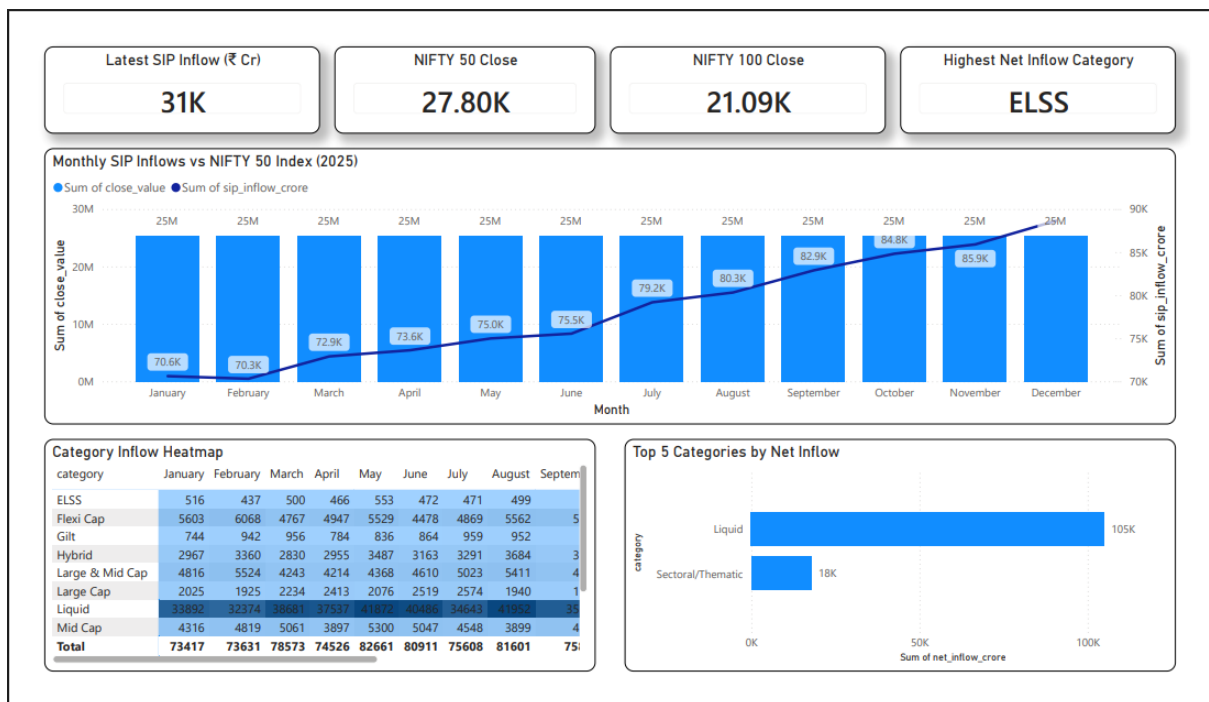
Page 4 – SIP & Market Trends

This page combines investment trends with market performance.

The dashboard includes:

- SIP Inflow vs NIFTY Trend
- Category-wise Net Inflow Heatmap
- Top Categories by Net Inflow
- Market Trend Analysis

This page allows users to compare mutual fund investments with broader market movements over time.



Monthly SIP Inflows vs NIFTY 50 Index (2025)

Sum of close_value Sum of sip_inflow_crore

Category Inflow Heatmap

category	January	February	March	April	May	June	July	August	Septem
ELSS	516	437	500	466	553	472	471	499	
Flexi Cap	5603	6068	4767	4947	5529	4478	4869	5562	5
Gilt	744	942	956	784	836	864	959	952	
Hybrid	2967	3360	2830	2955	3487	3163	3291	3684	3
Large & Mid Cap	4816	5524	4243	4214	4368	4610	5023	5411	4
Large Cap	2025	1925	2234	2413	2076	2519	2574	1940	1
Liquid	33892	32374	38681	37537	41872	40486	34643	41952	35
Mid Cap	4316	4819	5061	3897	5300	5047	4548	3899	4
Total	73417	73631	78573	74526	82661	80911	75608	81601	75

Top 5 Categories by Net Inflow

NAV Drill-through Page

A dedicated drill-through page was implemented to enable detailed analysis of individual mutual fund schemes.

The drill-through feature allows users to navigate directly from the Fund Performance table to a dedicated NAV history page while preserving the selected fund context.

The page includes:

- Historical NAV Trend
- Benchmark Comparison
- Fund Metadata
- Interactive Back Navigation



12.3 Dashboard Features

The Power BI dashboard incorporates several interactive features:

- KPI Cards
- Slicers
- Cross-filtering
- Drill-through Navigation
- Tooltips
- Dynamic Filtering
- Responsive Visualizations

These features improve usability and allow users to perform self-service analysis without modifying the underlying dataset.

13. Key Business Insights

The analytical findings generated during this project provide several valuable insights into the Indian mutual fund industry.

- The mutual fund industry experienced sustained growth in Assets Under Management throughout the analysis period, indicating increasing investor participation.
- Monthly SIP inflows showed a strong upward trend, reflecting growing confidence in systematic long-term investing.
- Equity-oriented mutual funds generally delivered stronger long-term returns than conservative investment categories.
- Risk-adjusted performance metrics identified several funds that consistently outperformed peers despite market volatility.
- Historical VaR and CVaR highlighted schemes with comparatively higher downside risk during adverse market conditions.
- Rolling Sharpe Ratios demonstrated that risk-adjusted performance changes significantly across different market cycles.
- Investor cohort analysis revealed that recent cohorts contributed a growing share of total investments.
- SIP continuity analysis identified a majority of investors maintaining regular investment schedules while also highlighting investors potentially at risk of discontinuation.
- Sector concentration analysis showed varying levels of portfolio diversification across equity funds.
- The Power BI dashboard enables interactive monitoring of industry performance, fund comparisons, investor trends, and benchmark analysis, supporting informed investment decisions.

14. Challenges Faced

Several technical and analytical challenges were encountered during the development of this project.

- Cleaning heterogeneous datasets collected from multiple sources.
- Standardizing inconsistent column names and data types.
- Handling missing values and duplicate records.
- Designing relationships across multiple datasets for Power BI.
- Computing rolling financial performance metrics efficiently.
- Implementing drill-through navigation and interactive filtering.
- Integrating statistical analysis with business intelligence reporting.
- Managing large historical NAV datasets while maintaining analytical performance.

Each challenge was addressed through systematic data preprocessing, modular Python scripts, database normalization, and iterative dashboard refinement.

15. Conclusion

This project successfully demonstrates an end-to-end data analytics solution for the Indian mutual fund industry. Beginning with raw datasets, a complete ETL pipeline was implemented to clean, standardize, and integrate financial data into a structured SQLite database. Comprehensive exploratory data analysis identified important trends related to fund performance, investor participation, SIP growth, and portfolio allocation.

Advanced financial analytics, including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR), provided a quantitative assessment of mutual fund performance and risk. Investor cohort analysis, SIP continuity analysis, and portfolio concentration metrics further enhanced the understanding of investor behavior and fund characteristics.

Finally, an interactive Power BI dashboard consolidated these analytical outputs into an intuitive decision-support platform. Overall, the project demonstrates the practical application of data engineering, statistical analysis, financial modeling, and business intelligence techniques to support data-driven investment analysis. Effective conclusions should summarize the work, reinforce the main findings, avoid introducing new information, and briefly point to future work.

16. Future Scope

Several opportunities exist to further enhance the project:

- Integrate live NAV data through the AMFI or MFAPI API.
- Automate daily ETL execution using scheduled workflows.
- Develop a Streamlit web application for online analytics.
- Implement Monte Carlo simulation for future NAV forecasting.
- Apply Markowitz portfolio optimization for optimal asset allocation.
- Incorporate machine learning models for mutual fund recommendation.
- Develop automated weekly performance reports delivered via email.
- Deploy the solution to a cloud platform for enterprise-scale usage.

17. References

Use a consistent citation style throughout the report (APA, IEEE, or your institution's preferred style). Ensure every source cited in the report appears here, and every reference listed is cited in the text.

Suggested references:

1. Python Software Foundation. *Python Documentation*.
<https://docs.python.org/>

2. Pandas Development Team. *Pandas Documentation*. <https://pandas.pydata.org/>
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5. Microsoft. *Power BI Documentation*. <https://learn.microsoft.com/power-bi/>
6. Plotly Technologies. *Plotly Python Documentation*. <https://plotly.com/python/>
7. Matplotlib Development Team. *Matplotlib Documentation*. <https://matplotlib.org/>
8. AMFI India. <https://www.amfiindia.com/>
9. National Stock Exchange of India. <https://www.nseindia.com/>
10. GitHub Documentation. <https://docs.github.com/>

18. Appendix

Appendix A – Project Folder Structure

```
bluestock_mf_capstone/  
├── data/  
│   ├── raw/      ← original downloaded files  
│   ├── processed/ ← cleaned, merged CSVs  
│   └── db/        ← bluestock_mf.db (SQLite)  
├── notebooks/  
│   ├── 01_data_ingestion.ipynb  
│   ├── 02_data_cleaning.ipynb  
│   ├── 03_eda_analysis.ipynb  
│   ├── 04_performance_analytics.ipynb  
│   └── 05_advanced_analytics.ipynb  
├── scripts/  
│   ├── etl_pipeline.py  
│   ├── live_nav_fetch.py  
│   ├── compute_metrics.py  
│   └── recommender.py  
├── sql/  
│   ├── schema.sql  
│   └── queries.sql  
├── dashboard/  
│   └── bluestock_mf.pbix  
├── reports/  
│   ├── Final_Report.pdf  
│   └── Presentation.pptx  
└── README.md
```

Appendix B – Sample SQL Queries

```
-- =====
-- Query 1: Top 5 Funds by AUM
-- =====

SELECT
    scheme_name,
    fund_house,
    aum_crore
FROM fact_performance
ORDER BY aum_crore DESC
LIMIT 5;

-- =====
-- Query 2: Average NAV Per Month
-- =====

SELECT
    strftime('%Y-%m', date) AS month,
    ROUND(AVG(nav), 2) AS avg_nav
FROM fact_nav
GROUP BY month
ORDER BY month;

-- =====
-- Query 3: SIP Year-over-Year Growth
-- =====

SELECT
    month,
    sip_inflow_crore,
    yoy_growth_pct
FROM monthly_sip_inflows
ORDER BY month;
```

Appendix C – GitHub Repository

https://github.com/kashyapkakadiya/bluestock_mf_capstone.git

